

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL24EQ001Ammonia compressor Interstage cooler-Stack.pvdb

Date of Analysis : Sep 8,2026 11:37am

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1060.000 mm ID	
Vessel Design Length, Tangent to Tangent	680.20 cm
Specified Datum Line Distance	0.00 cm
Shell Side Design Temperature	180 °C
Channel Side Design Temperature	85 °C
Shell Side Design Pressure	25.494 kgf/cm ²
Channel Side Design Pressure	20.395 kgf/cm ²
Shop Shell Side Test Pressure	38.867 kgf/cm ²
Field Shell Side Test Pressure	50.782 kgf/cm ²
Shop Channel Side Test Pressure	27.345 kgf/cm ²
Field Channel Side Test Pressure	27.383 kgf/cm ²
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 60	...	...	K02100	
Flange	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Re-Pad	SA-516 60	...	...	K02100	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-334 6	...	...	K03006	+IT
Tubesheet	SA-350 LF2	1	...	K03011	+IT
Flg Bolting	SA-320 L7	...	<= 65	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 60	...	...	K02100	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Blind Flg	20.520	0.00	21.000	3.0000	N/A	No
Ch Flg	20.500	0.00	21.100	3.0000	N/A	No
Ch Shell	20.501	0.00	28.600	3.0000	N/A	No
Main Shell	25.494	0.00	31.300	3.0000	N/A	No
Dish End	25.494	1.05	27.500	3.0000	No	No

Liquid Level: 125.00 cm Dens.: 4.998 kg/m³ Sp. Gr.: 0.005

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	9.8	9.5	103.0	95.0	88.0	...	1.00	1.00
Body Flg	23.1	13.3	113.0	105.0	96.6	...	1.00	1.00
Cylinder	93.1	70.0	14.0	14.0	10.8	...	1.00	1.00



Cylinder	675.1	570.0	15.0	15.0	12.8	...	1.00	1.00
Ellipse	680.2	5.1	19.0	15.0	14.1	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	95	57.7275	...	No Calc
20	30	105	86.4616	...	No Calc
30	Tsht	14	No Calc	...	14
Tsht	40	14	No Calc	...	No Calc
40	50	15	No Calc	...	3.77
50	60	15	5.82368	1.05461	11.5

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	83.297	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	583.833	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
T3-Tube Inlet	WRC-297	0.713	Shell	Passed
T4-Tube outlet	WRC-297	0.709	Shell	Passed
S1-Shell Inlet	WRC-297	0.814	Shell	Passed
S2-Shell outle	WRC-297	0.814	Shell	Passed

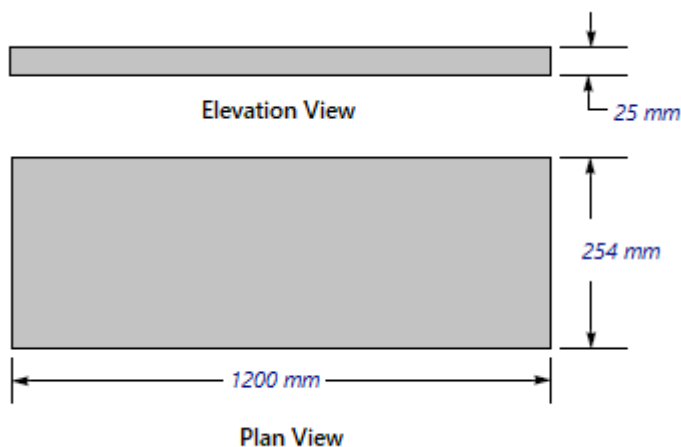
Foundation/Support Summary

Saddle Parameters:

Saddle Width	204.000	mm
Saddle Bearing Angle	155.000	deg.
Centerline Dimension	1000.000	mm
Wear Pad Width	350.000	mm
Wear Pad Thickness	14.000	mm
Wear Pad Bearing Angle	167.000	deg.
Distance from Saddle to Tangent	1400.000	mm

Baseplate Length	1200.000	mm
Baseplate Thickness	25.000	mm
Baseplate Width	254.000	mm
Number of Ribs (including outside ribs)	3	
Rib Thickness	14.000	mm
Web Thickness	14.000	mm
Height of Center Web	470.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Maximum Tensile Bolt Load 178. kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	36460.33	kgf
Maximum Transverse Saddle Shear Load	4797.33	kgf
Maximum Longitudinal Saddle Shear Load	3350.94	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	23676.95	kgf
Maximum Transverse Saddle Shear Load	152.34	kgf
Maximum Longitudinal Saddle Shear Load	78.94	kgf

Weights:

Fabricated - Bare w/o removable internals	24802.7	kg
Shop Test - Fabricated + water (full)	30318.1	kg
Shipping - Fab. + rem. intls.+ shipping app.	25253.3	kg
Erected - Fab. + rem. intls.+ insul. (etc)	25253.3	kg
Empty - Fab. + intls. + details + wghts.	25253.3	kg
Operating - Empty + operating liquid (no ca)	28727.3	kg
Field Test - Empty weight + test medium (full)	30768.7	kg



Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
T3-Tube Inlet	28.6	...	...	OK	10.16	OK	Passed
T4-Tube outlet	28.6	...	...	OK	10.16	OK	Passed
S1-Shell Inlet	29.9	...	...	OK	11.33	OK	Passed
S2-Shell outlet	29.9	...	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 29.9 Nozzle : S2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 28.6 Nozzle : T3-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	T3-Tube Inlet	581.175	270.000	358.100
30	T4-Tube outlet	581.175	90.000	358.100
40	S1-Shell Inlet	6051.175	90.000	1021.701
40	S2-Shell outlet	1751.175	270.000	1021.701

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
T3-Tube Inlet	200 mm	160	WNF	219.1	23.012	319.08	16.00	325	300
T4-Tube outlet	200 mm	160	WNF	219.1	23.012	319.08	16.00	325	300
S1-Shell Inlet	600 mm	140	WNF	609.6	52.375	...	...	449	300
S2-Shell outlet	600 mm	140	WNF	609.6	52.375	...	...	449	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
T3-Tube Inl	SA-333 6	14.000	10.000	12.000	16.000	...
T4-Tube out	SA-333 6	14.000	10.000	12.000	16.000	...
S1-Shell In	SA-350 LF2	15.000	10.000	...	...	...
S2-Shell ou	SA-350 LF2	15.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
T3-Tube Inlet	58.118	270.0	300.00	0.00	Ch Shell
T4-Tube outlet	58.118	90.0	300.00	0.00	Ch Shell
S1-Shell Inlet	605.117	90.0	300.00	0.00	Main Shell
S2-Shell outlet	175.117	270.0	300.00	0.00	Main Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 103.0mm THK X 1250.0mm OD	SA-350 LF2
1	WELD NECK FLANGE: 113.0mm THK X 1250.0mm OD	SA-350 LF2
1	ELLIPTICAL HEAD: 2.0 X 1, 19.0mm THK X 1050.0mm ID X 5.1cm	SA-516 60
5	BAFFLES: 1050.0mm DIA X 10.0mm THK	SA-516 70
10	TIE RODS: 25mm DIA X 5000mm	SA-193 B7M
1	TUBESHEET: 1092.0mm X 120.0mm THK	SA-350 LF2
396	TUBES: 5250.0mm X 25.4mm DIA X 2.8mm THK	SA-334 6
2	SADDLE: 204.0mm X 155 DEG	SA-516 60
1	INSULATION: 5700mm X 100mm THK	Personal protec
1	INSULATION: 313mm X 100mm THK	Personal protec
1	GASKET: 1114.0mm OD X 1082.0mm ID	...
32	BODY FLANGE BOLTS: 27.0mm DIA	SA-320 L7
64	NUTS FOR BODY FLANGE BOLTS: 27.0mm DIA	...
1	CYLINDRICAL SEGMENT 14.0mm THK X 3077.0mm X 70.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 253.1mm X 70.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 15.0mm THK X 249.8mm X 570.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 15.0mm THK X 3080.3mm X 570.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Main Shell [8]	!		-35	-37		0.956	15.000	1.00	Yes
S1-Shell Inlet[1]	!		-46	-46		0.814	15.000	1.00	Yes
Nozzle Flg [5]	A		-18	-48					
S2-Shell outle[1]	!		-46	-46		0.814	15.000	1.00	Yes
Nozzle Flg [5]	A		-18	-48					
Tubesheet: SS[13]	!		-46	-46		0.594	30.000	1.00	Yes
Warmest MDMT:			-18	-37					
Blind Flg [11]	!		-46	-46		0.994	23.750	1.00	No
Ch Flg [11]	!		-46	-46		0.733	14.000	1.00	No
Ch Shell [8]	!		0	-15		0.731	14.000	1.00	No
Dish End [10]	C		-34	-47	-29	0.765	15.000	1.00	No
Dish End [7]	C		-28	-48	-29	0.583	19.000	1.00	No
T3-Tube Inlet [1]	C		-46	-46		0.712	16.000	1.00	No
Nozzle Flg [5]	A		-18	-48					
T4-Tube outlet[1]	C		-46	-46		0.708	16.000	1.00	No
Nozzle Flg [5]	A		-18	-48					
Tubesheet: CS[14]	!		-46	-46		0.493	30.000	1.00	No
Bolting [21]			-101						
Warmest MDMT:			0	-15					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail			
Shell		-37.0		-35.0		Pass			
Channel/Tube		-15.0		0.0		Pass			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.





Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc/m2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	25.493	kgf/cm ²
Channel Side Design Pressure	20.395	kgf/cm ²
Shell Side Design Temperature	180.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-35.0	°C
MDMT (CET), Tube Side	0.0	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 cm
Distance Upwind of Crest	Lh 0 cm
Distance from Crest to the Vessel	x 0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
--	-----



Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Blind Flg
Distance "FROM" to "TO"	9.5 cm
Flange Outside Diameter	1250 mm
Element Thickness	95 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	103 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	20.395 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2294.2 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2340.9 kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	1 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	125 cm
Liquid Density	999.54 kg/m ³

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Ch Flg
Distance "FROM" to "TO"	13.3 cm
Flange Inside Diameter	1050 mm
Element Thickness	105 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	113 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	20.395 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²



Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	105 cm
Liquid Density	999.54 kg/m ³

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Ch Shell
Distance "FROM" to "TO"	70 cm
Inside Diameter	1060 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	20.395 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2386 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2467.6 kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	106 cm
Liquid Density	999.54 kg/m ³

Element From Node	30
Detail Type	Nozzle
Detail ID	T3-Tube Inlet
Dist. from "FROM" Node / Offset dist	35 cm
Nozzle Diameter	200 mm
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	59.638 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT

Element From Node	30
Detail Type	Nozzle
Detail ID	T4-Tube outlet
Dist. from "FROM" Node / Offset dist	35 cm



Nozzle Diameter	200	mm
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.048	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6	+IT

Element#	4/5	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	570	cm
Inside Diameter	1060	mm
Element Thickness	15	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	15	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.494	kgf/cm ²
Design Temperature Internal Pressure	180	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	180	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	140	cm
Width of Saddle	204	mm
Height of Saddle at Bottom	1000	mm
Saddle Contact Angle	155.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	350	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	167.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40	
Detail Type	Saddle	
Detail ID	RHS Saddle	
Dist. from "FROM" Node / Offset dist	430	cm
Width of Saddle	204	mm
Height of Saddle at Bottom	1000	mm
Saddle Contact Angle	155.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	350	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	167.0	
Friction coefficient	7.9999998e-002	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40
Detail Type	Liquid



Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	106	cm
Liquid Density	4.9977	kg/m ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Personal protec	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	570	cm
Thickness of Insulation	100	mm
Density	0.0001918	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	S1-Shell Inlet	
Dist. from "FROM" Node / Offset dist	500	cm
Nozzle Diameter	600	mm
Nozzle Schedule	140	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	613.94	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	S2-Shell outlet	
Dist. from "FROM" Node / Offset dist	70	cm
Nozzle Diameter	600	mm
Nozzle Schedule	140	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	613.94	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	40	
Detail Type	Weight	
Detail ID	Datasheet weigh	
Dist. from "FROM" Node / Offset dist	250	cm
Miscellaneous Weight	399.99	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	TOP HE EMPTY	
Dist. from "FROM" Node / Offset dist	200	cm
Miscellaneous Weight	12380	kgf
Offset from Element Centerline	2000	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	TOP HE OPERATIN	
Dist. from "FROM" Node / Offset dist	200	cm
Miscellaneous Weight	1970	kgf
Offset from Element Centerline	2000	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	TOP HE TEST	
Dist. from "FROM" Node / Offset dist	200	cm
Miscellaneous Weight	5965.9	kgf
Offset from Element Centerline	2000	mm
Element From Node	40	
Detail Type	For./Mom.	
Detail ID	TOP HE FORCES	



Dist. from "FROM" Node / Offset dist	0	cm
Force in X Direction	1386	kgf
Force in Y Direction	0	kgf
Force in Z Direction	359.99	kgf
Moment about X Axis	0	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	0	kgf-m.
Force/Moment Combination Method	SRSS	

Element From Node	40	
Detail Type	For./Mom.	
Detail ID	TOP HE FORCES	
Dist. from "FROM" Node / Offset dist	0	cm
Force in X Direction	1965	kgf
Force in Y Direction	0	kgf
Force in Z Direction	1965	kgf
Moment about X Axis	0	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	0	kgf-m.
Force/Moment Combination Method	SRSS	

Element#	5/5	
Element From Node	50	
Element To Node	60	
Element Type	Elliptical	
Description	Dish End	
Distance "FROM" to "TO"	5.08	cm
Inside Diameter	1050	mm
Element Thickness	15	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	19	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.494	kgf/cm ²
Design Temperature Internal Pressure	180	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	180	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 60	
Allowable Stress, Ambient	1203.2	kgf/cm ²
Allowable Stress, Operating	1203.2	kgf/cm ²
Allowable Stress, Hydrotest	2018.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	1951.7	kgf/cm ²
UCS-66 Chart Curve Designation	C	
External Pressure Chart Name	CS-2	
UNS Number	K02100	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	105	cm
Liquid Density	4.9977	kg/m ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Personal protec	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	31.33	cm
Thickness of Insulation	100	mm
Density	0.0001918	kg/cm ³

PV Elite 28 Licensee: SL User
FileName : A150NRL24EQ001_Ammonia compressor Interstage coo Page 18 of 145
Input Echo: Step: 1 11:37am Sep 8,2026



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XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Blind Flg		9.8175	...	9.5	...
Ch Flg		23.1175	...	13.3	...
Ch Shell		93.1175	...	70	...
Main Shell		675.117	...	570	...
Dish End		680.198	...	5.08	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

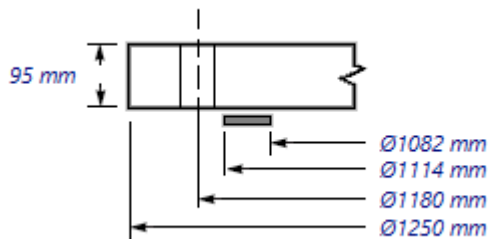
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Flange Input Data Values

Description: Blind Flg :

Blind Flg

Flange Type	TEMA Channel Cover		
Design Pressure	P	20.52	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1250.000	mm
Flange Thickness	t	95.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	3707.079	mm
Youngs Modulus for Blind Flange Matl.	Sy	2029713.75	kgf/cm ²
Allowed Channel Cover Deflection	dfAllow	1.3125	mm
Diameter of Bolt Circle	C	1180.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1125.000	mm
Flange Face Inside Diameter	Fid	1050.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1114.000	mm
Gasket Inside Diameter	Gi	1082.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	16.000 mm
Basic Gasket Width,	$bo = N/2$	8.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1099.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1100^2 \cdot 20.5$

= 194915.078 kgf

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1100 \cdot 3 \cdot 20.5$$

$$= 30317.707 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(194915 + 30318 + 0, 0)$$

$$= 225232.781 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 7.13 \cdot \pi \cdot 1100 + 0 \cdot 0 \cdot 0$$

$$= 173129.719 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((225233 + 0 + 4 \cdot 0/1100)/1754, 173130/1754)$$

$$= 128.424 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	128.424	132.433	
Radial Distance between Bolts and Edge:	30.500	35.000	
Circ. Spacing between the Bolts:	65.000	115.660	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 132 \cdot 1754 / (703 \cdot \pi (1114 + 1082))$$

$$= 4.788 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1754 \cdot 132$$

$$= 232262.64 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 225232.78 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1180 - 1100) / 2$$

$$= 40.1274 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 225233 \cdot 40.1 \cdot 1 / (92^2 \cdot 1100) + 0.3 \cdot 1 \cdot 20.5 \cdot 1100^2 / 92^2$$

$$= 1064.12 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 232263 \cdot 40.1 \cdot 1 / (92^2 \cdot 1100)$$

$$= 190.24 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 225233 / 132$$

$$= 1700.74 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:



$$= (Wm^2 / Ab)$$

$$= (173130/132)$$

$$= 1307.31 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	1064.12	1407.13	190.24	1407.13
Bolting	1700.74	1753.82	1307.31	1753.82

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 1.1504 mm
 Allowed deflection at center of Cover 1.3125 mm

Reqd. Blind Flange Thickness at Center 83.005 mm
 Reqd. Blind Flange Thickness at Gasket 36.828 mm
 Min. Required Flange Thickness per TEMA 88.044 mm

Estimated M.A.W.P. (Operating) 21.2 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 903.6 kg
 Estimated Unfinished Weight of Forging at given Thk 903.6 kg

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	997.97	1407.13	178.42	1407.13
Bolting	1700.74	1753.82	1307.31	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 87.528 mm

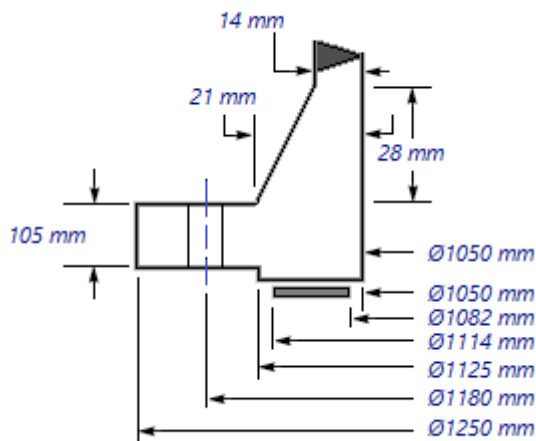
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Flange Input Data Values

Description: FLANGE :

Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	20.50	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1050.000	mm
Flange Outside Diameter	A	1250.000	mm
Flange Thickness	t	105.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	28.0000	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material	SA-320 L7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	1180.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads	TEMA Metric		
Number of Bolts	32		
Flange Face Outside Diameter	Fod	1125.000	mm
Flange Face Inside Diameter	Fid	1050.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1114.000	mm
Gasket Inside Diameter	Gi	1082.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (20.5 \cdot (1050/2 + 3)) / (1407 \cdot 1 - 0.6 \cdot 20.5) + Ca$$

= 10.7600 mm

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 11) / (528 + 0.6 \cdot 11)$$

$$= 28.953 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 14) / (525 + 0.6 \cdot 14)$$

$$= 36.933 \text{ kgf/cm}^2$$

Corroded Flange Thickness, tc = T-ci	102.000 mm
Corroded Flange ID, Bcor = B+2*ci	1056.000 mm
Corroded Large Hub, g1Cor = g1-ci	18.000 mm
Corroded Small Hub, g0Cor = go-ci	11.000 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	44.000 mm
Gasket Contact Width, N = (Go - Gi)/2	16.000 mm
Basic Gasket Width, bo = N/2	8.000 mm
Effective Gasket Width, b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter, G = Go - 2 * b	1099.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1100^2 \cdot 20.5$$

$$= 194725.172 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1100 \cdot 3 \cdot 20.5$$

$$= 30288.170 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1056^2 \cdot 20.5$$

$$= 179541.938 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 194725 - 179542$$

$$= 15183.241 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H_p, 0)$$

$$= \max(194725 + 30288 + 0, 0)$$

$$= 225013.344 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 7.13 \cdot \pi \cdot 1100 + 0 \cdot 0 \cdot 0$$

$$= 173129.719 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a)$$

$$= \max((225013 + 0 + 4 \cdot 0 / 1100) / 1754, 173130 / 1754)$$

$$= 128.299 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 27 + 6 \cdot 102 / (3 + 0.5)$$

$$= 228.857 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1180 \cdot \sin(\pi / 32)$$

$$= 115.660 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{116/(2 \cdot 27 + 102)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	128.299	132.433	
Radial Distance between Hub and Bolts:	38.000	44.000	
Radial Distance between Bolts and Edge:	30.500	35.000	
Circ. Spacing between the Bolts:	65.000	115.660	228.857

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 132 \cdot 1754 / (703 \cdot \pi(1114 + 1082))$$

$$= 4.788 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1754 \cdot 132$$

$$= 232262.64 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 225013 - 194725$$

$$= 30288.17 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1180 - 1100) / 2$$

$$= 40.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1180 - 1056) / 2 + 40.1)$$

$$= 51.0637 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1180 - 1056 - 18) / 2$$

$$= 52.9999 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	179542.	52.9999	1.0000	9516.
Face Pressure, Mt	15183.	51.0637	1.0000	775.
Gasket Load, Mg	30288.	40.1274	1.0000	1215.
Gasket Seating, Matm	232263.	40.1274	1.0000	9320.

Total Moment for Operation, Mo 11506. kgf-m.
 Total Moment for Gasket seating, Mg 9320. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 107.778 mm
 Hub Ratio, h/h0 = HL / H0 0.260
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.636

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.886
 Factor V per 4.16.5 0.364

Factor f per 4.16.5 1.544

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1250/1056$$

$$= 1.184$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.18^2 \cdot \log_{10}(1.18))) / (1.18^2 - 1)) / (1.18 - 1)$$

$$= 11.600$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.18^2(1 + 8.55246 \cdot \log_{10}(1.18)) - 1) / ((1.0472 + 1.9448 \cdot 1.18^2) \cdot (1.18 - 1))$$

$$= 1.845$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.18^2(1 + 8.55246 \cdot \log_{10}(1.18)) - 1) / (1.36136(1.18^2 - 1) \cdot (1.18 - 1))$$

$$= 12.747$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.18^2 + 1) / (1.18^2 - 1)$$

$$= 5.985$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 12.7 \cdot 11^2 \cdot 108 / 0.36$$

$$= 456111.781 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.89/108$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (102 \cdot 0.0082 + 1) / 1.85 + 102^3/456112$$

$$= 3.323$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.54 \cdot 11506/1056) / (3.32 \cdot 18^2)$$

$$= 1562.69 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.54 \cdot 9320/1056) / (3.32 \cdot 18^2)$$

$$= 1265.77 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.12 \cdot 11506/1056) / (3.32 \cdot 102^2)$$

$$= 66.76 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.12 \cdot 9320/1056) / (3.32 \cdot 102^2)$$

$$= 54.08 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (11.6 \cdot 11506 / (102^2 \cdot 1056)) - 5.99 \cdot 66.8$$

$$= 815.24 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (11.6 \cdot 9320 / (102^2 \cdot 1056)) - 5.99 \cdot 54.1$$

$$= 660.34 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1563 + \max(66.8, 815)) / 2$$

$$= 1188.97 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (1266 + \max(54.1, 660)) / 2$$

$$= 963.05 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 225013 / 132$$

$$= 1699.08 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (173130 / 132)$$

$$= 1307.31 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed	
Longitudinal Hub	1562.69	2110.70	1265.77	2110.70
Radial Flange	66.76	1407.13	54.08	1407.13
Tangential Flange	815.24	1407.13	660.34	1407.13
Maximum Average	1188.97	1407.13	963.05	1407.13
Bolting	1699.08	1753.82	1307.31	1753.82

Minimum Required Flange Thickness [Rigidity] 96.571 mm

Estimated M.A.W.P. (Operating) 21.2 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 306.7 kg
 Estimated Unfinished Weight of Forging at given Thk 372.4 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 9320 / 1 \cdot 100000 \cdot 0.36 / (2.75 \cdot 2067025 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.795 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 11506 / 1 \cdot 100000 \cdot 0.36 / (2.75 \cdot 2029713 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.999 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 9320 / 1 \cdot 100000 \cdot 0.36 / (3.32 \cdot 2067025 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.659 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 11506 / 1 \cdot 100000 \cdot 0.36 / (3.32 \cdot 2029713 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.828 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	1401.95	2110.70	1107.82 2110.70
Radial Flange	81.71	1407.13	64.57 1407.13
Tangential Flange	670.69	1407.13	529.98 1407.13
Maximum Average	1036.32	1407.13	818.90 1407.13
Bolting	1699.08	1753.82	1307.31 1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 85.877 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Blind Flg		20.52	103	3	1250	1407.1	1
Ch Flg		20.5	113	3	1050	1407.1	2
Ch Shell		20.501	14	3	1060	1407.1	3
Main Shell		25.494	15	3	1060	1407.1	4
Dish End		25.494	19	3	1050	1203.2	5

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Blind Flg		20.3948	21	21.2	95	88.0436	1
Ch Flg		20.3948	21.1	21.2	105	96.5708	2
Ch Shell		20.3948	28.6	36.6	14	10.834	3
Main Shell		25.4935	31.3	39.2	15	12.7629	4
Dish End		25.4935	27.5	34.3	15	14.1271	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

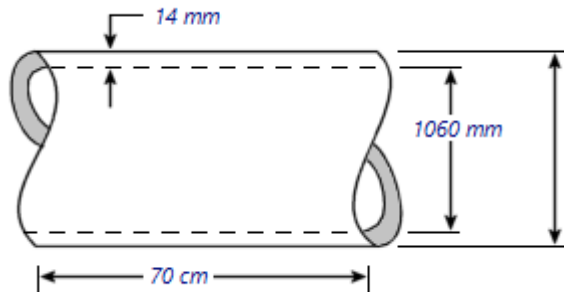
Shell Side MAWP = 29.9 kgf/cm²
 Shell Side MAPnc = 39.2 kgf/cm²
 Channel Side MAWP = 21 kgf/cm², < design pressure: 25.494 kgf/cm²
 Channel Side MAPnc = 21.2 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	20.501 kgf/cm²	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		70.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1060.000 mm	Actual Stress	1005.7 kgf/cm²	
Spec. Min. Thk t _e , or sa		14.0000 mm	Spec. Nominal Thk.	14.000 mm	
Surface Area		23926.4 cm²	SG of Contents	1.000	
Element Volume		617732.8 cm³	Weight of Contents	617.5 kgf	
Empty Weight		256.3 kgf	Operating Weight	873.7 kgf	
			Comp. All. Ope. Stress	1330.1 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (20.5 \cdot 533) / (1407 \cdot 1 - 0.6 \cdot 20.5)$$

$$= 7.8340 + 3.0000 = 10.8340 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.106 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 11) / (533 + 0.6 \cdot 11)$$

$$= 28.7 - 0.11 = 28.6 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 14) / (530 + 0.6 \cdot 14)$$

$$= 36.6 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (20.5 \cdot (533 + 0.6 \cdot 11)) / (1 \cdot 11)$$

$$= 1005.670 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.304 %

Minimum Design Metal Temperature Results:



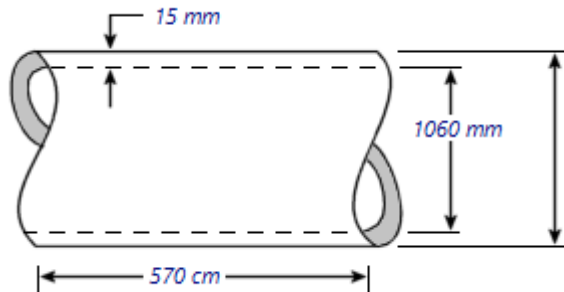
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	25.494 kgf/cm²	Int. Design Temperature	180 °C	
Material		SA-516 70	Ext. Design Temperature	180 °C	
Length		570.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1060.000 mm	Actual Stress	1147.7 kgf/cm²	
Spec. Min. Thk t _e , or s _a		15.0000 mm	Spec. Nominal Thk.	15.000 mm	
Surface Area		195187.2 cm²	SG of Contents	0.005	
Element Volume		3837776.8 cm³	Weight of Contents	19.2 kgf	
Empty Weight		2237.9 kgf	Operating Weight	2257.1 kgf	
			Comp. All. Ope. Stress	1259.0 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.5 \cdot 533) / (1407 \cdot 1 - 0.6 \cdot 25.5)$$

$$= 9.7629 + 3.0000 = 12.7629 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 12) / (533 + 0.6 \cdot 12)$$

$$= 31.3 - 0.00053 = 31.3 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 15) / (530 + 0.6 \cdot 15)$$

$$= 39.2 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.5 \cdot (533 + 0.6 \cdot 12)) / (1 \cdot 12)$$

$$= 1147.656 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.395 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Note:

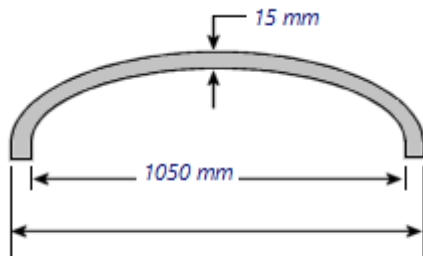
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Dish End

Elliptical Head Design Information:							
Design Pressure	P	25.494	kgf/cm²	Int. Design Temperature		180	°C
Material		SA-516	60	Ext. Design Temperature		180	°C
Length		5.080	cm	External Pressure Chart		CS-2	
				UNS Number		K02100	
Int. Corr. All.	c	3.0000	mm	Allowable Stress (ope)	S	1203.2	kgf/cm²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1203.2	kgf/cm²
Inside Diameter	Di	1050.000	mm	Actual Stress		1115.9	kgf/cm²
Spec. Min. Thk t _e , or s _a		15.0000	mm	Spec. Nominal Thk.		19.000	mm
Surface Area		14568.0	cm²	SG of Contents		0.005	
Element Volume		195521.0	cm³	Weight of Contents		1.0	kgf
Empty Weight		226.5	kgf	Operating Weight		227.5	kgf
				Comp. All. Ope. Stress		1259.0	kgf/cm²
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.5 \cdot 1056 \cdot 0.99) / (2 \cdot 1203 \cdot 1 - 0.2 \cdot 25.5)$$

$$= 11.1271 + 3.0000 = 14.1271 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1203 \cdot 1 \cdot 12) / (0.99 \cdot 1056 + 0.2 \cdot 12)$$

$$= 27.5 - 0.00053 = 27.5 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1203 \cdot 1 \cdot 15) / (1 \cdot 1050 + 0.2 \cdot 15)$$

$$= 34.3 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.5 \cdot (0.99 \cdot 1056 + 0.2 \cdot 12)) / (2 \cdot 1 \cdot 12)$$

$$= 1115.861 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (25.5 \cdot 528) / (1203 \cdot 1 - 0.6 \cdot 25.5) + 3$
 $= 14.332 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.001 kgf/cm²
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1203 \cdot 1 \cdot 16) / (528 + 0.6 \cdot 16)$
 $= 35.8 - 0.00053 = 35.8 \text{ kgf/cm}^2$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1056 / (2 \cdot 266))^2) / 6$
 $= 0.992488$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 7.580 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 15$, $t_r = 9.18$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.76$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -34 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -47 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 19$, $t_r = 9.33$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.58$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -28 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	21.035 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	15.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	0.765	Temperature Difference	13 °C
UCS-66(c) red.	No	Computed Min. Temp.	-47 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

EXTERNAL PRESSURE RESULTS for: Dish End

Elliptical Head From 50 to 60 Ext. Chart: CS-2 at 180 °C

Elastic Modulus from Chart: CS-2 at 180 °C: 1999551 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
12.000	1080.00	90.00	0.0015432	930.67

$MAEP = B / (K_0 \cdot Do / t) = 931 / (0.9 \cdot 90) = 11.5 \text{ kgf/cm}^2$



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.824	1080.00	382.48	0.0003631	363.05

$$MAEP = B/(K0*Do/t) = 363/(0.9 *382) = 1.05 \text{ kgf/cm}^2$$

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	38.867 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	33.142 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	50.782 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	32.888 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	36.456 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	36.456 kgf/cm ²
App 27-4	= M.A.W.P.	29.898 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	27.345 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	33.142 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	27.383 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	23.138 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	36.456 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	36.456 kgf/cm ²
App 27-4	= M.A.W.P.	21.035 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Ch Shell	1055.7	2386.0	0.442	27.45
Main Shell	1400.4	2386.0	0.587	38.97
Dish End	963.5	2018.9	0.477	27.45

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T3-Tube Inlet	Nozzle	1203.20	1203.20	1.000
T3-Tube Inlet	Pad	1203.20	1203.20	1.000
T4-Tube outlet	Nozzle	1203.20	1203.20	1.000
T4-Tube outlet	Pad	1203.20	1203.20	1.000
S1-Shell Inlet	Nozzle	1407.13	1407.13	1.000
S2-Shell outlet	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Blind Flg	1407.13	1407.13	1.000
Ch Flg	1407.13	1407.13	1.000
Ch Shell	1407.13	1407.13	1.000
Main Shell	1407.13	1407.13	1.000
Dish End	1203.20	1203.20	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1203.20	1203.20	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
T3-Tube Inlet	119.68	1564.16	0.077
T4-Tube outlet	119.68	1564.16	0.077
S1-Shell Inlet	211.22	1829.27	0.115
S2-Shell outlet	211.22	1829.27	0.115

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	38.867 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	33.142 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	50.782 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	32.888 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	36.456 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	36.456 kgf/cm ²
App 27-4	= M.A.W.P.	29.898 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	27.345 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	33.142 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	27.383 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	23.138 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	36.456 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	36.456 kgf/cm ²
App 27-4	= M.A.W.P.	21.035 kgf/cm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Ch Shell	1348.5	2386.0	0.565	27.49
Main Shell	2290.8	2386.0	0.960	50.89
Dish End	1203.1	2018.9	0.596	27.49

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T3-Tube Inlet	Nozzle	1203.20	1203.20	1.000
T3-Tube Inlet	Pad	1203.20	1203.20	1.000
T4-Tube outlet	Nozzle	1203.20	1203.20	1.000
T4-Tube outlet	Pad	1203.20	1203.20	1.000
S1-Shell Inlet	Nozzle	1407.13	1407.13	1.000
S2-Shell outlet	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Blind Flg	1407.13	1407.13	1.000
Ch Flg	1407.13	1407.13	1.000
Ch Shell	1407.13	1407.13	1.000
Main Shell	1407.13	1407.13	1.000
Dish End	1203.20	1203.20	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1203.20	1203.20	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
T3-Tube Inlet	119.85	1564.16	0.077
T4-Tube outlet	119.85	1564.16	0.077
S1-Shell Inlet	275.79	1829.27	0.151
S2-Shell outlet	275.79	1829.27	0.151

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		50.78 kgf/cm ²
Shell Thickness	ts	15.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1060.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.13 kgf/cm ²
Shell Allowable Stress at Ambient		1407.13 kgf/cm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		27.38 kgf/cm ²
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1060.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	396
Tube Wall Thickness	et	2.7700 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5250.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5130.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		1203.20 kgf/cm ²
Tube Allowable Stress At Ambient		1203.20 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2447.19 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	40.5000 mm
Tube Layout Pattern		Square
Fillet Weld Leg	af	2.7700 mm
Groove Weld Leg	ag	2.7700 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	517.036 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm
Area of Tube Layout	Ap	6133.6 cm ²
Perimeter of Tube Layout	Cp	3085.1 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a



Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-350 LF2

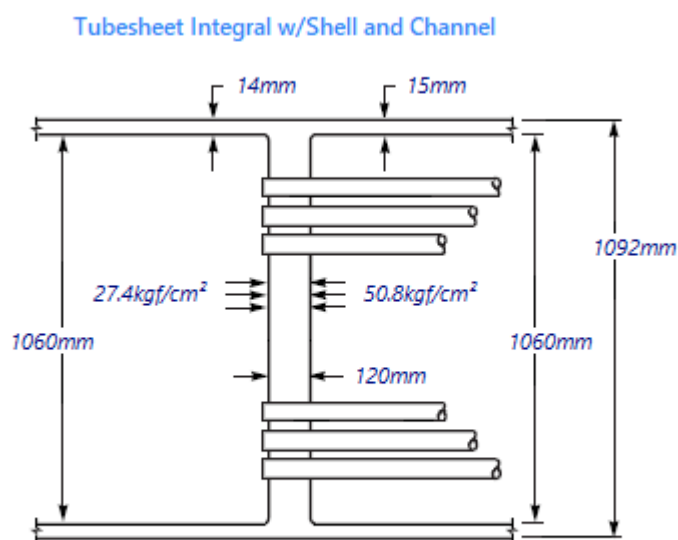
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 120.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
 Tubesheet Outside Diameter A 1092.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20670E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (27.4 \cdot (25.4/2 - 0)) / (1203 \cdot 1 + 0.4 \cdot 27.4)$
 $= 0.2864 + 0.0000 = 0.2864 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	25.40	5647.04	9.17	50.0000	0.0130823	1251.46

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 192 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.9655	25.40	5647.04	26.31	50.0000	0.0015894	1002.05

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 1002) / (3 \cdot 26.3) = 50.8 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.9655 mm
Allowable Internal Pressure at Corroded thickness	287.51 kgf/cm ²
Required Internal Design Pressure	27.38 kgf/cm ²
Allowable External Pressure at Corroded thickness	191.50 kgf/cm ²
Required External Design Pressure	50.78 kgf/cm ²
Required Thickness due to Shell Side pressure	0.9655 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot r_o + dt$
 $= 2 \cdot 517 + 25.4$
 $= 1059.472 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (40.5 - 25.4) / 40.5$
 $= 0.373$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= D_s / D_o$ (Configurations a, b, c)
 $= 1060 / 1059$
 $= 1.0005$

Ratio [pc]:

$= D_c / D_o$ (Configurations a, e, f)
 $= 1060 / 1059$
 $= 1.0005$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 0/120 \\
 &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned}
 &= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\
 &= \max(25.4 - 2 \cdot 2.77 (2067025/2067025) \cdot \\
 &\quad (1203/1407) \cdot (0); 25.4 - 2 \cdot 2.77) \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\
 &= 40.5 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1059 \cdot 40.5) / (\pi \cdot 1059^2)) } \\
 &= 40.5000 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (40.5 - 25.4) / 40.5 \\
 &= 0.3728
 \end{aligned}$$

$$\begin{aligned}
 h/p &= 2.962963; \mu^* = 0.372840; E^* = 880911 \text{ kgf/cm}^2 \\
 E^*/E &= 0.426173; \nu^* = 0.319366
 \end{aligned}$$

Note: As h/p (2.963) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1060 + 15) 15) } \\
 &= 0.0143 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\
 &= 0.014 \cdot 2067025 \cdot 15^3 / (6(1 - 0.3^2)) \\
 &= 18290806.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(1060 + 15) \cdot 18290806 / 120^3 \cdot (1 + 120 \cdot 0.014 + (120 \cdot 0.014)^2 / 2) \\
 &= 286294.1 \text{ kgf/cm}^2
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\
 &= 1060^2 / (4 \cdot 2067025 \cdot 15) \cdot (1 - 1060 \cdot 0.3 / (2(1060 + 15))) \\
 &= 0.77197E-02 \text{ mm/kgf/cm}^2
 \end{aligned}$$

Parameter [ω_s]:

$$\begin{aligned}
 &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\
 &= 1 \cdot 18290806 \cdot 0.014 \cdot 0.00771972 (1 + 120 \cdot 0.014) \\
 &= 5496.4604 \text{ mm}^2
 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned}
 &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1060 + 14) 14) } \\
 &= 0.0148 \text{ 1/mm}
 \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned}
 &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\
 &= 0.015 \cdot 2067025 \cdot 14^3 / (6(1 - 0.3^2)) \\
 &= 15400217.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned}
 &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\
 &\quad h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\
 &= 6(1060 + 14 / \cos(0)) \cdot 15400217 / 120^3 \cdot (1 + \\
 &\quad 120 \cdot 0.015 \cdot (1 + 1) / 2 + (120 \cdot 0.015)^2 / 2 \cdot 1)
 \end{aligned}$$

$$= 250472.8 \text{ kgf/cm}^2$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1060^2 / (4 \cdot 2067025 \cdot 14) \cdot (1 - 1060 \cdot 0.3 / (2(1060 + 14)))$$

$$= 0.82698E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta c (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1 \cdot 15400217 [0.008269792 (1 + 0.015 \cdot 1 \cdot 120) - 0.000 (1/120 + 1 \cdot 0.015/2)]$$

$$= 5249.4331 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1092/1059$$

$$= 1.0307$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 880911 \cdot (286294 + 250473 + 2067025 \cdot \ln(1.03))$$

$$= 0.4630$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [M_t]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1059^2 / 16 \cdot [(1 - 1) \cdot (1^2 + 1) \cdot 50.8 - (1 - 1) \cdot (1^2 + 1) \cdot 27.4]$$

$$= 1636.7762 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M^*]:

$$= M_t + \omega c \cdot P_t - \omega s \cdot P_s$$

$$= 1637 + 5249 \cdot 27.4 - 5496 \cdot 50.8$$

$$= -133739.4062 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [M_p]:

$$= (M^*) - D_o^2 / 32 \cdot F (P_s - P_t) / (1 + F)$$

$$= (-133739) - 1059^2 / 32 \cdot 0.46 (50.8 - 27.4) / (1 + 0.46)$$

$$= -351177.9375 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= -351178 + 1059^2 / 64 \cdot (3 + 0.32) \cdot (50.8 - 27.4)$$

$$= 1011056.0625 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|-351178|; |1011056|)$$

$$= 1011056.0625 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1011056 / (0.37 (120 - 6)^2)$$

$$= 1251.9724 \text{ kgf/cm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 2549$$

$$= 4843.40 \text{ kgf/cm}^2$$

Tubesheet bending stress at final thickness $[\sigma]$:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 783650 / (0.37 \cdot (57.1 - 6)^2)$$

$$= 4836.2573 \text{ kgf/cm}^2$$

Required tubesheet thickness for Bending Stress $[H_{reqB}]$:

$$= h + C_{ats} + C_{atc}$$

$$= 57.1 + 0 + 0$$

$$= 57.0647 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) $[H_{req}]$:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(57.1; 6.44)$$

$$= 57.0647 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential $[|P_s - P_t|]$:

$$= |P_s - P_t|$$

$$= |50.8 - 27.4|$$

$$= 23.399 \text{ kgf/cm}^2$$

Shear Stress check $[\tau_{limit}]$:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta_T \cdot S_y$$

$$= 4 \cdot 0.37 \cdot 120 / 1059 \cdot 0.8 \cdot 0.95 \cdot 2549$$

$$= 327.253 \text{ kgf/cm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}) . Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear $[H_{reqS}]$:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta_T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.37 \cdot 1937) \cdot (4 \cdot 6134 / 3085) \cdot |50.8 - 27.4| + 0$$

$$= 6.4403 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 15.000mm

Min. Shell len. adjacent to the tubesheet is: 226.971 mm

Note: For a given Channel thickness of: 14.000 mm

Min. Channel len. adjacent to the tubesheet is: 219.275mm

Tube Metal Area $[A_t]$:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength $[F_t]$:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 2325$$

$$= 4578.307 \text{ kgf}$$

Groove Weld Strength $[F_g]$:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 2325$$

$$= 4578.3066 \text{ kgf}$$

Weld Strength Factor $[f_w]$:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 2325 / 2325$$

$$= 1.000$$

Design Strength Ratio $[f_d]$:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 2325$$

$$= 3032.7986 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77)1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 3.372 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm

Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 4578 is $\geq$ tube strength 4578 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	1252.0 $\leq$	4843.4 kgf/cm ²	Ok
Tubesheet shear stress	104.0 $\leq$	1937.4 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	57.065	120.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case	Pass/
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	62.509	...	1478	4843	122	1937	Ps+Pt D1	Ok
D2uc	88.371	...	2730	4843	226	1937	Ps+Pt D2	Ok
D3uc	57.065	...	1252	4843	104	1937	Ps+Pt D3	Ok
Max:	88.3710	... mm		0.564		0.116	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	3778	...	3778	Ok
D2uc	...	...	3778	...	3778	Ok



D3uc	...	...	...	3778	...	3778	Ok
Max	...	...	...	...	...	...	...

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	88.371	120.000	Ok
Tube thickness :	0.966	2.770	Ok
Tube-tubesheet fillet weld leg :	1.686	2.770	Ok
Tube-tubesheet groove weld leg :	1.686	2.770	Ok

Given Shell Thickness: 15.0000 mm
 Given Channel Thickness: 14.0000 mm

Min Shell length of thk, (15.000) adj. to tubesheet: 226.971 mm

Min Channel length of thk, (14.000) adj. to tubesheet: 219.275 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	92	No Calc	No Calc
20	30	No Calc	...	102	No Calc	No Calc
30	Tsht	83.297	1088	11	0.0018179	1035.51
Tsht	40	No Calc	...	11	No Calc	No Calc
40	50	583.833	1090	12	0.00025671	256.656
50	60	No Calc	1080	12	0.0015432	930.67

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	95	57.7275	...	No Calc
20	30	105	86.4616	...	No Calc
30	Tsht	14	No Calc	...	14
Tsht	40	14	No Calc	...	No Calc
40	50	15	No Calc	...	3.77
50	60	15	5.82368	1.05461	11.5

Minimum

3.77

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	83.297	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	583.833	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	979.647	...	979.647	...	97.9647
20	30	329.145	123214	318.3	123492	32.9145
30	40	256.273	617733	201.92	624746	25.6273
40	50	2237.93	3837776	1795.34	3894882	223.793
50	60	226.529	195521	190.761	198638	22.6529
Total		4029	4774244.50	3485	4841757.00	402



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	...	4.75	...	...	Liquid: 10
20	Liqd	123.158	6.65	...	...	Liquid: 20
30	Liqd	617.454	35	...	...	Liquid: 30
30	Nozl	65.6013	35	-61.6525	...	T3-Tube Inlet
30	Nozl	81.4527	35	61.6525	...	T4-Tube outlet
40	Sadl	261.316	140	75.05	...	Left Saddle
40	Sadl	261.316	430	75.05	...	RHS Saddle
40	Liqd	19.1802	285	...	...	Liquid: 40
40	Insl	409.146	285	...	...	Personal protec
40	Nozl	675.334	500	78.2425	...	S1-Shell Inlet
40	Nozl	675.334	70	-78.2425	...	S2-Shell outlet
40	Wght	399.993	250	...	...	Datasheet weigh
40	Wght	12379.8	200	200	...	TOP HE EMPTY
40	Wght	1969.97	200	200	...	TOP HE OPERATIN
40	Wght	5965.9	200	200	...	TOP HE TEST
40	Forc	...	...	...	...	TOP HE FORCES
40	Forc	...	...	...	...	TOP HE FORCES
50	Liqd	0.97716	13.83	0.36335E-05	...	Liquid: 50
50	Insl	41.4522	15.665	...	...	Personal protec
30	FTsh	752.864	76	...	...	
30	Tube	3840.23	332.5	...	...	Tubes
40	Bafl	195.15	54	...	...	Baffles
40	Bafl	195.15	147.6	...	...	Baffles
40	Bafl	195.15	241.2	...	...	Baffles
40	Bafl	195.15	334.8	...	...	Baffles
40	Bafl	196.397	428.4	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	522.6
Liquid	760.8
Insulation	450.6
Nozzles	1497.7
Weights	20715.6
Exchanger Components	5570.1
Liquid in Tubes	743.3
Sum of the Detail Weights	30260.8 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
--	------------	-----------	----------	---------	-------	-----------



Main Elements	4432.5	4432.5	4432.5	4432.5	4432.5	4432.5
Saddles	522.6	522.6	522.6	522.6	522.6	522.6
Nozzles	1497.7	1497.7	1497.7	1497.7	1497.7	1497.7
Wld Weights	12779.8	12779.8	12779.8	12779.8	12779.8	12779.8
Exchanger	5570.1	5570.1	5570.1	5570.1	5570.1	5570.1
Insulation	...	...	450.6	450.6	450.6	450.6
Ope Weights	...	...	...	...	...	1970.0
Ope. Liquid	...	...	...	...	...	760.8
Tube Ope Lqd	...	...	...	...	...	743.3
Test Liquid	...	4772.1	...	...	...	...
Tube Tst Lqd	...	743.3	...	...	...	...
Totals	24802.7	30318.1	25253.3	25253.3	25253.3	28727.3

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	24802.7 kg
Shop Test Wt.	- Fabricated weight + water (full)	30318.1 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	25253.3 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	25253.3 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	25253.3 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	28727.3 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	28129.4 kg
Field Test Wt.	- Empty weight + test medium (full)	30768.7 kg

Exchanger Tube Data

Volume of Exchanger tubes :	743655.6 cm ³
Weight of Ope Liq in tubes :	743.3 kg
Weight of Water in tubes :	743.3 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	16316.6	...
20	30	9010.91	4651.13
30	40	23926.4	23310.6
40	50	195187	189815
50	60	14568	13626.7

Total	259009.1	231403.5 cm ²
Total	278.8	249.1 ft ²

Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
T3-Tube Inlet	48.57	52.01	85	300	GR 1.1	3.200	48.571	48.571	48.571
T4-Tube outlet	48.57	52.01	85	300	GR 1.1	3.200	48.571	48.571	48.571
S1-Shell Inlet	45.20	52.01	180	300	GR 1.1	1.097	45.195	45.195	45.195
S2-Shell outle	45.20	52.01	180	300	GR 1.1	1.097	45.195	45.195	45.195

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 45.195 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 52.007 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 48.571 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 52.007 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
T3-Tube Inlet	248.539	0.50	1	1530	21.035	48.571	Pass
T4-Tube outlet	248.539	0.50	1	1530	21.035	48.571	Pass
S1-Shell Inlet	657.225	0.50	8	3671	29.898	45.195	Pass
S2-Shell outlet	657.225	0.50	8	3671	29.898	45.195	Pass

Evaluating Flange on Nozzle [T3-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(1) + 4 \cdot \text{abs}(1530) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot [(48.571 - 21.035) + 500 \cdot 48.571]$$

$$= 0.15432E+04 \leq 0.24995E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle T3-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.571, 0.500 \cdot 48.571 + 48.571 - 1543.2 / (\pi \cdot 248.539^3))$$

$$= \min(48.571, 69.657)$$

$$= 48.571 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [T4-Tube outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(1) + 4 \cdot \text{abs}(1530) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot [(48.571 - 21.035) + 500 \cdot 48.571]$$

$$= 0.15432E+04 \leq 0.24995E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle T4-Tube outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.571, 0.500 \cdot 48.571 + 48.571 - 1543.2 / (\pi \cdot 248.539^3))$$

$$= \min(48.571, 69.657)$$

$$= 48.571 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [S1-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(8) + 4 \cdot \text{abs}(3671) \cdot 657.225 \leq \pi \cdot 657.225^3 \cdot [(45.195 - 29.898) + 500 \cdot 45.195]$$

$$= 0.97807E+04 \leq 0.33796E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle S1-Shell Inlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(45.195, 0.500 \cdot 45.195 + 45.195 - 9780.7 / (\pi \cdot 657.225^3)) \\
 &= \min(45.195, 66.696) \\
 &= 45.195 \text{ kgf/cm}^2
 \end{aligned}$$

Evaluating Flange on Nozzle [S2-Shell outlet]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(8) + 4 \cdot \text{abs}(3671) 657.225 \leq \pi \cdot 657.225^3 [(45.195 - 29.898) + 0.500 \cdot 45.195] \\
 &= 0.97807E+04 \leq 0.33796E+06 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle S2-Shell outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(45.195, 0.500 \cdot 45.195 + 45.195 - 9780.7 / (\pi \cdot 657.225^3)) \\
 &= \min(45.195, 66.696) \\
 &= 45.195 \text{ kgf/cm}^2
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.99) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.92)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.581
Structure Height to Diameter ratio	5.846

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5 : Z_g = 58979 \text{ cm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 100 + 0 = 100 \text{ cm}$
 $= 3.28 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (3.28 ft.) $< 15 \text{ ft.}$
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15/1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[K_{zt}]$:
 $K_1 = 0, K_2 = 0, K_3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [162 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.58 \cdot 1.53$$

$$= 25.1 \text{ lbs. } [11.4 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Blind Flg	100.0	0.000	0.000	0.000	1.035	1.000	161.847
Ch Flg	100.0	0.000	0.000	0.000	1.035	1.000	161.847
Ch Shell	100.0	0.000	0.000	0.000	1.035	1.000	161.847
Main Shell	100.0	0.000	0.000	0.000	1.035	1.000	161.847
Dish End	100.0	0.000	0.000	0.000	1.035	1.000	161.847

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Datasheet wei	0.00	100.00	161.85	0.00
TOP HE EMPTY	1.00	300.00	166.07	0.02
TOP HE OPERAT	1.00	300.00	166.07	0.02
TOP HE TEST	1.00	300.00	166.07	0.02

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	100	150	1425	161.847	12.5236
20	30	100	126	1675.8	161.847	14.7278
30	40	100	130.56	9139.2	161.847	80.3198
40	50	100	154.8	88236	161.847	775.516
50	60	100	153.6	4575.33	161.847	40.2103

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(5.330.75)$
 = 0.070 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.07$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 27607 \cdot 1.5$
 = 2699.946 kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 27607$
 = 1200.0 kgf

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	62.5	3943.83	444.655
20	30	52.5	3943.83	373.51
30	40	53	3943.83	377.068
40	Sad1	53	3943.83	377.068
Sad1	50	53	3943.83	377.068
40	50	53	3943.83	377.068



50	60	52.5	3943.83	373.51
----	----	------	---------	--------

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	390.117 cm
Center of Gravity of the Liquid	60.555 cm
Center of Gravity of the Insulation	417.777 cm
Center of Gravity of the Nozzles	357.520 cm
Center of Gravity of the Added Weights (Operating)	306.473 cm
Center of Gravity of the Added Weights (Empty)	306.682 cm
Center of Gravity of the Tubesheet(s)	99.118 cm
Center of Gravity of the Tubes	355.617 cm
Center of Gravity of the Baffles	346.556 cm
Center of Gravity of Bare Shell New and Cold	261.789 cm
Center of Gravity of Bare Shell Corroded	245.030 cm
Vessel CG in the Operating Condition	301.352 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	308.323 cm
Vessel CG in the Test Condition	315.905 cm

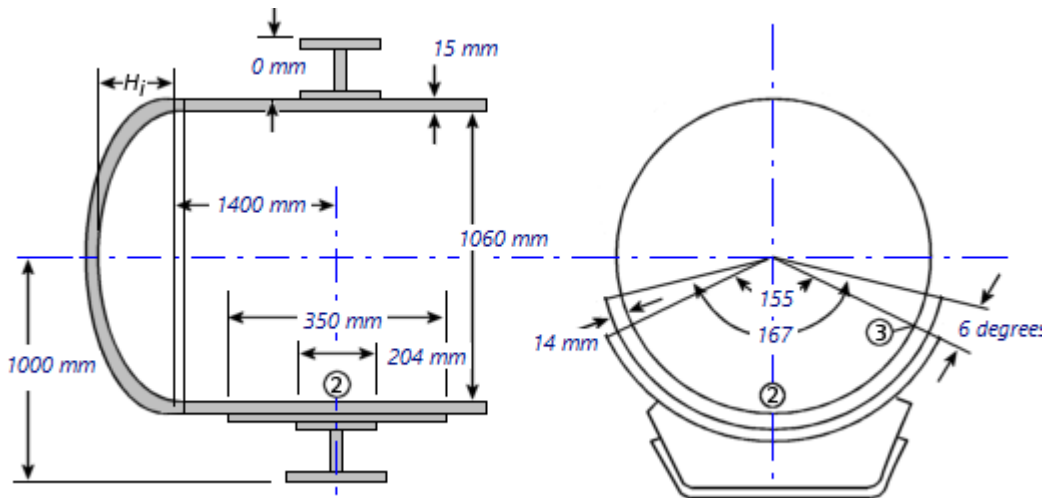
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	539.00	mm
Shell Thickness used in this Case	t	15.000	mm
Stiffened Vessel Length per 4.15.6	L	570.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1400.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	155.00	degrees
Wear Plate Width	b2 or b1	350.00	mm
Wear Plate Bearing Angle	delta2 or theta1	167.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	36199.02	kgf
Pressure used in Analysis	P	25.494 kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm²	Allowable kgf/cm²
Long. Stress at Top of Midspan	555.86	1407.13
Long. Stress at Bottom of Midspan	589.23	1407.13
Long. Stress at Top of Saddles	1187.88	1407.13
Long. Stress at Bottom of Saddles	216.02	1407.13
Tangential Shear in Shell	214.78	1125.70
Circ. Stress at Horn of Saddle	337.49	1758.91
Circ. Compressive Stress in Shell	28.02	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (923/2 + 3447) \cdot 1000/1091$$

$$= 10743.9 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(239, 3351) \cdot 1000/2900$$

$$= 1155.5 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(2700, 3351) \cdot 1000/2900$$

$$= 1155.5 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (2700/2 + 3447) \cdot 1000/1091$$

$$= 13185.5 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 23013 + \max(1155, 10744, 1155, 13186)$$

$$= 36199.0 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4(1.09 \cdot 1.2)^2)$$

$$= 2345.980 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	36460.33	kgf
Transverse Shear Load	4797.33	kgf
Longitudinal Shear Load	3350.94	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1060/2 + 15 + 14) \sin(155/2), 2 \cdot 539)$$

$$= 1091.499 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1710	K2 = 0.7543	K3 = 0.4386	K4 = 0.2805
K5 = 0.6630	K6 = 0.0289	K7 = 0.0289	K8 = 0.2942
K9 = 0.2072	K10 = 0.0316	K1* = 0.2951	K6p = 0.0229
K7p = 0.0229			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L} \cdot \text{a})) / (1 + (4\text{hm})/3\text{L})]$$

$$= -36199 \cdot 140 [1 - (1 - 140/570 + (53.9^2 - 0^2)/(\text{L} \cdot \text{a})) / (1 + (4 \cdot 0)/(3 \cdot 570))]$$

$$= -11524.9 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot \text{L} / 4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - 4\text{a}/\text{L})$$

$$= 36199 \cdot 570 / 4 ((1 + 2(53.9^2 - 0^2)/(570^2)) / (1 + (4 \cdot 0)/(3 \cdot 570)) - 4 \cdot 140/570)$$

$$= 1827.5 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$$

$$= 25.5 \cdot 539/(2 \cdot 12) - 1827/(\pi \cdot 539^2 \cdot 12)$$

$$= 555.86 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$$



$$= 25.5 \cdot 539 / (2 \cdot 12) + 1827 / (\pi \cdot 539^2 \cdot 12)$$

$$= 589.23 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 25.5 \cdot 539 / (2 \cdot 12) - 11525 / (0.17 \cdot \pi \cdot 539^2 \cdot 12)$$

$$= 1187.88 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 25.5 \cdot 539 / (2 \cdot 12) + 11525 / (0.3 \cdot \pi \cdot 539^2 \cdot 12)$$

$$= 216.02 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 36199 (570 - 2 \cdot 140) / (570 + (4 \cdot 0 / 3))$$

$$= 18417.0 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.75 \cdot 18417 / (539 \cdot 12)$$

$$= 214.78 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(539 \cdot 12)}$$

$$= 62.731 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(539 \cdot 12)}, 2 \cdot 140)$$

$$= 329.46 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1407 / 1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.66 \cdot 36199 \cdot 0.1 / (329(12 + 1 \cdot 14))$$

$$= -28.02 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -36199 / (4(12 + 1 \cdot 14)329) -$$

$$3 \cdot 0.029 \cdot 36199 / (2(12 + 1 \cdot 14)^2)$$

$$= -337.49 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 2900.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12620\text{E-}04 \cdot 2900 (180 - 21.1)$$

$$= 5.815 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	1951.6	kgf/cm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(533 \cdot 12)} + 350)^3) / 12$$

$$= 6.837 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 6.84 + 57 (192 - 6)^2$$

$$= 19754.939 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	474.8	12.0	6.0	57.0	34182.8	0.198E+05
Wearplate	350.0	14.0	19.0	49.0	93100.0	0.147E+05
Web	14.0	416.0	234.0	58.2	1362815.4	0.942E+04
BasePlate	254.0	25.0	454.5	63.5	2886074.2	0.437E+05
Totals	...	...	...	227.7	4376172.5	0.876E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4376172 / 228$$

$$= 192.181 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 155 / 2$$

$$= 102.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(102) - 0.5 \cdot \sin(102)^2) / (\pi - 1.79 + \sin(102) \cos(102))$$

$$= 0.2690$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.27 \cdot 36199$$

$$= 9736.5225 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 57.0255 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1170.9894 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000 - 533 \cdot \sin(155/2) / 1.35$$

$$= 615.293 \text{ mm}$$

Bending Moment, M = Fh * d = 5990.8242 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 131.4225 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1301.0994 kgf/cm²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(36199 + 261)254 / (4 \cdot 1200 \cdot 1407)}$$

$$= 20.282 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 155/2) (530 + 15 + 14)$$

$$= 1091.499 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1091 / (3 - 1)$$

$$= 545.749 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 546 \cdot 254 / 2$$

$$= 693.102 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 36460 / (1200 \cdot 254)$$

$$= 11.962 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 693 \cdot 12$$

$$= 8290.919 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 26.6 + 38.2$$

$$= 64.802 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 8291 / 64.8$$

$$= 127.942 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	204.0	...	28.6	...	990.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 14$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 64.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204 / 2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(990 / 64.8)}$$

$$= 39.095 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(155 / 2) (530 + 15 + 14) - 25$$

$$= 854.010 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 1952)}$$

$$= 143.602$$

Slenderness ratio [KL/r]:

$$= \text{KL} / \text{r}$$

$$= 1 \cdot 854/39.1$$

$$= 21.844$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 3351 / (2 \cdot 1200) \cdot 546 \cdot 854 / 2$$

$$= 325.374 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($21.8 < 144$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (21.8)^2 / (2 \cdot 144^2)) 1952 /$$

$$(5/3 + 3 \cdot (21.8) / (8 \cdot 144) - (21.8^3) / (8 \cdot 144^3))$$

$$= 1119 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 128/1119 + (325 \cdot 102/9904609) / 1301$$

$$= 0.372$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	190.0	0.0	26.6	0.0	990.
Web	545.7	14.0	0.0	76.4	0.0	12.5
Totals	...	...	...	103.0	...	0.100E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/103$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1000 - \cos(559/2) (25 + 0 + 0) - 0$$

$$= 416.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1003/103) }$$

$$= 31.199 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 416/31.2$$

$$= 13.334$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3351 / (2 \cdot 1200)$$

$$= 1.396 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.4 \cdot 546 \cdot 416$$

$$= 316.988 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σB ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 317/98296$$

$$= 322.483 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($13.3 < 144$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$



$$= (1 - (13.3)^2 / (2 \cdot 144^2)) 1952 / (5/3 + 3 \cdot (13.3) / (8 \cdot 144) - (13.3^3) / (8 \cdot 144^3))$$

$$= 1142 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 156/1142 + (317 \cdot 102/1003) / 1301$$

$$= 0.384$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1529.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	1203.21 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load QO (Weight)	QO	23274.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1155.5 kgf
Maximum Transverse Force	Ft	4797.3 kgf
Maximum Longitudinal Force	Fl	3350.9 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 484.4 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = max(Ff, Fl) / (Bolt Area • Number of Bolts)
 = 3351 / (4.87 • 4)
 = 172.2 kgf/cm². Must be less than 1019.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = Ft / (Bolt Area • Number of Bolts)
 = 4797 / (4.87 • 4)
 = 246.5 kgf/cm². Must be less than 1019.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 3351 / (1019 • 4)
 = 0.8221 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1000 • 4797 + 0
 = 4797.33 kgf-m.

Eccentricity (e):
 = Rmom / QO
 = 4797/23275
 = 206.12 mm > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis
 = 1200/2 - 50.8
 = 549.20 mm

Modular Ratio Of Steel/Concrete (n1):
 = ES / EC
 = 2074056/219500
 = 9.45



$$\begin{aligned} K1 &= 3 (e - 0.5 \cdot Bplen) \\ &= 3 (206 - 0.5 \cdot 1200) \\ &= -1181.65 \text{ mm} \end{aligned}$$

$$\begin{aligned} K2 &= 6 \cdot n1 \cdot At / Bpwid \cdot (f + e) \\ &= 6 \cdot 9.45 \cdot 9.73/254 \cdot (549 + 206) \\ &= 164079.75 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} K3 &= -K2 \cdot (0.5 \cdot Bplen + f) \\ &= -164080 \cdot (0.5 \cdot 1200 + 549) \\ &= -188560437.43 \text{ mm}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\ Y^3 + -1182 \cdot Y^2 + 164080 \cdot Y + -188560432 &= 0 \\ Y &= 1178.22 \text{ mm} \end{aligned}$$

$$\begin{aligned} Num &= (Bplen / 2 - Y / 3 - e) \\ &= (1200/2 - 1178/3 - 206) \\ &= 1.14 \end{aligned}$$

$$\begin{aligned} Denom &= (Bplen / 2 - Y / 3 + f) \\ &= (1200/2 - 1178/3 + 549) \\ &= 756.46 \end{aligned}$$

Total Bolt Tension Force [Tforce]:

$$\begin{aligned} &= -QO \cdot Num / Denom \\ &= -23275 \cdot 1.14/756 \\ &= -35.18 \text{ kgf} \end{aligned}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= Tforce / (Stba \cdot Nbt) \\ &= -35.2 / (1529 \cdot 2) \\ &= -0.0115 \text{ cm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0, 0.82, -0.012] \\ &= 0.8221 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned} &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(23275 + -35.2) / (1178 \cdot 254) \\ &= 15.53 \text{ kgf/cm}^2 \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$\begin{aligned} &= (Bplen - WebIngt) / 2 \\ &= (1200 - 1149) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

Overturning Moment due To Bolt Tension [Mt]:

$$\begin{aligned} &= Tforce \cdot Adist \\ &= -35.2 \cdot 2.54 \\ &= -0.89 \text{ kgf-m.} \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= fc \cdot (Y - Adist) / Y \\ &= 15.5 \cdot (1178 - 25.4) / 1178 \\ &= 15.20 \text{ kgf/cm}^2 \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned} &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 254/6) \cdot (15.2 + 2 \cdot 15.5) \\ &= 12.63 \text{ kgf-m.} \end{aligned}$$

Baseplate Required Thickness [Treq]:



$$= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$$

$$= \sqrt{(6 \cdot \max(-0.89, 12.6) / (254 \cdot 1805))}$$

$$= 4.0664 \text{ mm}$$

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	539.00	mm
Shell Thickness used in this Case	t	15.000	mm
Stiffened Vessel Length per 4.15.6	L	570.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1400.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	155.00	degrees
Wear Plate Width	b2 or b1	350.00	mm
Wear Plate Bearing Angle	delta2 or theta1	167.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	27.15	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1203.20	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		12588.62	kgf
Pressure used in Analysis	P	25.494	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		577.40	1407.13
Long. Stress at Bottom of Midspan		567.70	1407.13
Long. Stress at Top of Saddles		834.04	1407.13
Long. Stress at Bottom of Saddles		421.04	1407.13
Tangential Shear in Shell		70.23	1125.70
Circ. Stress at Horn of Saddle		117.37	1758.91
Circ. Compressive Stress in Shell		9.74	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (923/2 + 1122) • 1000/1091
 = 4353.8 kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(239, 3351) • 1000/2900
 = 1155.5 kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 = max(Fl,eq; Sum of X Forces) • B / Ls
 = max(2700, 3351) • 1000/2900
 = 1155.5 kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:



$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (2700/2 + 1122) \cdot 1000/1091$$

$$= 6795.3 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 5793 + \max(1155, 4354, 1155, 6795)$$

$$= 12588.6 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4(1.09 \cdot 1.2)^2)$$

$$= 2345.980 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	12849.94	kgf
Transverse Shear Load	2472.37	kgf
Longitudinal Shear Load	3350.94	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1060/2 + 15 + 14)\sin(155/2), 2 \cdot 539)$$

$$= 1091.499 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1710	K2 = 0.7543	K3 = 0.4386	K4 = 0.2805
K5 = 0.6630	K6 = 0.0289	K7 = 0.0289	K8 = 0.2942
K9 = 0.2072	K10 = 0.0316	K1* = 0.2951	K6p = 0.0229
K7p = 0.0229			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$$

$$= -12589 \cdot 140 [1 - (1 - 140/570 + (53.9^2 - 27.2^2)/(2 \cdot 140 \cdot 570))/(1 + (4 \cdot 27.2)/(3 \cdot 570))]$$

$$= -4897.6 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$$

$$= 12589 \cdot 570/4((1 + 2(53.9^2 - 27.2^2)/(570^2))/(1 + (4 \cdot 27.2)/(3 \cdot 570)) - 4 \cdot 140/570)$$

$$= -531.4 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 25.5 \cdot 539/(2 \cdot 12) - 531/(\pi \cdot 539^2 \cdot 12)$$

$$= 577.40 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 25.5 \cdot 539/(2 \cdot 12) + 531/(\pi \cdot 539^2 \cdot 12)$$

$$= 567.70 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot \text{Rm}/(2t) - M1/(K1 \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 25.5 \cdot 539/(2 \cdot 12) - 4898/(0.17 \cdot \pi \cdot 539^2 \cdot 12)$$

$$= 834.04 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot \text{Rm}/(2t) + M1/(K1^* \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 25.5 \cdot 539/(2 \cdot 12) + 4898/(0.3 \cdot \pi \cdot 539^2 \cdot 12)$$

$$= 421.04 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot \text{hm}/3))$$

$$= 12589 (570 - 2 \cdot 140) / (570 + (4 \cdot 27.2/3))$$

$$= 6022.3 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.75 \cdot 6022 / (539 \cdot 12)$$

$$= 70.23 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(539 \cdot 12)}$$

$$= 62.731 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(539 \cdot 12)}, 2 \cdot 140)$$

$$= 329.46 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1407/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.66 \cdot 12589 \cdot 0.1 / (329(12 + 1 \cdot 14))$$

$$= -9.74 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -12589 / (4(12 + 1 \cdot 14)329) -$$

$$3 \cdot 0.029 \cdot 12589 / (2(12 + 1 \cdot 14)^2)$$

$$= -117.37 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	1951.6	kgf/cm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(533 \cdot 12)} + 350)12^3) / 12$$

$$= 6.837 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 6.84 + 57(192 - 6)^2$$

$$= 19754.939 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	474.8	12.0	6.0	57.0	34182.8	0.198E+05
Wearplate	350.0	14.0	19.0	49.0	93100.0	0.147E+05
Web	14.0	416.0	234.0	58.2	1362815.4	0.942E+04
BasePlate	254.0	25.0	454.5	63.5	2886074.2	0.437E+05
Totals	...	...	...	227.7	4376172.5	0.876E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4376172/228$$

$$= 192.181 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 155/2$$

$$= 102.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(102) - 0.5 \cdot \sin(102)^2) / (\pi - 1.79 + \sin(102) \cos(102))$$

$$= 0.2690$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.27 \cdot 12589$$

$$= 3385.9875 \text{ kgf}$$

Tension Stress, $St = (Fh/As) = 19.8313 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 1170.9894 \text{ kgf/cm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 533 \cdot \sin(155/2) / 1.35$$

$$= 615.293 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 2083.3779 \text{ kgf-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 45.7037 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 1301.0994 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(12589 + 261) \cdot 254 / (4 \cdot 1200 \cdot 1407)}$$

$$= 12.040 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 155/2) (530 + 15 + 14)$$

$$= 1091.499 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1091 / (3 - 1)$$

$$= 545.749 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 546 \cdot 254 / 2$$

$$= 693.102 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 12850 / (1200 \cdot 254)$$

$$= 4.216 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 693 \cdot 4.22$$

$$= 2922.020 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 26.6 + 38.2$$

$$= 64.802 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 2922/64.8$$

$$= 45.091 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	204.0	...	28.6	...	990.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 14$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/64.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(990/64.8)}$$

$$= 39.095 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(155/2) (530 + 15 + 14) - 25$$

$$= 854.010 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/1952)}$$

$$= 143.602$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 854/39.1$$

$$= 21.844$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 3351 / (2 \cdot 1200) \cdot 546 \cdot 854/2$$

$$= 325.374 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($21.8 < 144$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (21.8)^2 / (2 \cdot 144^2)) 1952 /$$

$$(5/3 + 3 \cdot (21.8) / (8 \cdot 144) - (21.8)^3 / (8 \cdot 144^3))$$

$$= 1119 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 45.1/1119 + (325 \cdot 102/9904609) / 1301$$

$$= 0.298$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction



	B	D	Y	A	AY	Io
Rib	14.0	190.0	0.0	26.6	0.0	990.
Web	545.7	14.0	0.0	76.4	0.0	12.5
Totals	...	...	...	103.0	...	0.100E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/103$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(559/2)(25 + 0 + 0) - 0$$

$$= 416.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1003/103}$$

$$= 31.199 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 416/31.2$$

$$= 13.334$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3351/(2 \cdot 1200)$$

$$= 1.396 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.4 \cdot 546 \cdot 416$$

$$= 316.988 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 317/98296$$

$$= 322.483 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($13.3 < 144$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.3)^2 / (2 \cdot 144^2)) 1952 / (5/3 + 3 \cdot (13.3) / (8 \cdot 144) - (13.3)^3 / (8 \cdot 144^3))$$

$$= 1142 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 54.9/1142 + (317 \cdot 102/1003) / 1301$$

$$= 0.296$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1529.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	1203.21 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	6054.6 kgf



Saddle Load QL (Wind/Seismic contribution)	QL	1155.5 kgf
Maximum Transverse Force	Ft	2472.4 kgf
Maximum Longitudinal Force	Fl	3350.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = $Fl / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $3351 / (4.87 \cdot 4)$
 = $172.2\ kgf/cm^2$. Must be less than $1019.0\ kgf/cm^2$.

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $2472 / (4.87 \cdot 4)$
 = $127.0\ kgf/cm^2$. Must be less than $1019.0\ kgf/cm^2$.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $Fl / (BoltShearAllowable \cdot Nbolts)$
 = $3351 / (1019 \cdot 4)$
 = $0.8221\ cm^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + Sum\ of\ X\ Moments$
 = $1000 \cdot 2472 + 0$
 = $2472.37\ kgf-m$.

Eccentricity (e):
 = $Rmom / QO$
 = $2472 / 6055$
 = $408.35\ mm > Bplen / 6$ --> Uplift in Transverse direction

$f = Bplen / 2 - Edgedis$
 = $1200 / 2 - 50.8$
 = $549.20\ mm$

Modular Ratio Of Steel/Concrete (n1):
 = ES / EC
 = $2074056 / 219500$
 = 9.45

$K1 = 3 (e - 0.5 \cdot Bplen)$
 = $3 (408 - 0.5 \cdot 1200)$
 = $-574.96\ mm$

$K2 = 6 \cdot n1 \cdot At / Bpwid \cdot (f + e)$
 = $6 \cdot 9.45 \cdot 9.73 / 254 \cdot (549 + 408)$
 = $208010.42\ mm^2$

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$
 = $-208010 \cdot (0.5 \cdot 1200 + 549)$
 = $-239045564.51\ mm^3$

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -575 \cdot Y^2 + 208010 \cdot Y + -239045568 = 0$
 $Y = 734.70\ mm$

$Num = (Bplen / 2 - Y / 3 - e)$
 = $(1200 / 2 - 735 / 3 - 408)$
 = -53.24

$Denom = (Bplen / 2 - Y / 3 + f)$
 = $(1200 / 2 - 735 / 3 + 549)$
 = 904.30



Total Bolt Tension Force [Tforce]:

$$\begin{aligned} &= - QO \cdot Num / Denom \\ &= - 6055 \cdot -53.2/904 \\ &= 356.49 \text{ kgf} \end{aligned}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= Tforce / (Stba \cdot Nbt) \\ &= 356 / (1529 \cdot 2) \\ &= 0.1166 \text{ cm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0, 0.82, 0.12] \\ &= 0.8221 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned} &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(6055 + 356) / (735 \cdot 254) \\ &= 6.87 \text{ kgf/cm}^2 \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$\begin{aligned} &= (Bplen - WebIngh) / 2 \\ &= (1200 - 1149) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

Overturning Moment due To Bolt Tension [Mt]:

$$\begin{aligned} &= Tforce \cdot Adist \\ &= 356 \cdot 2.54 \\ &= 9.05 \text{ kgf-m.} \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= fc \cdot (Y - Adist) / Y \\ &= 6.87 \cdot (735 - 25.4) / 735 \\ &= 6.63 \text{ kgf/cm}^2 \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned} &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 254 / 6) \cdot (6.63 + 2 \cdot 6.87) \\ &= 5.56 \text{ kgf-m.} \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned} &= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))} \\ &= \sqrt{(6 \cdot \max(9.05, 5.56) / (254 \cdot 1805))} \\ &= 3.4426 \text{ mm} \end{aligned}$$

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

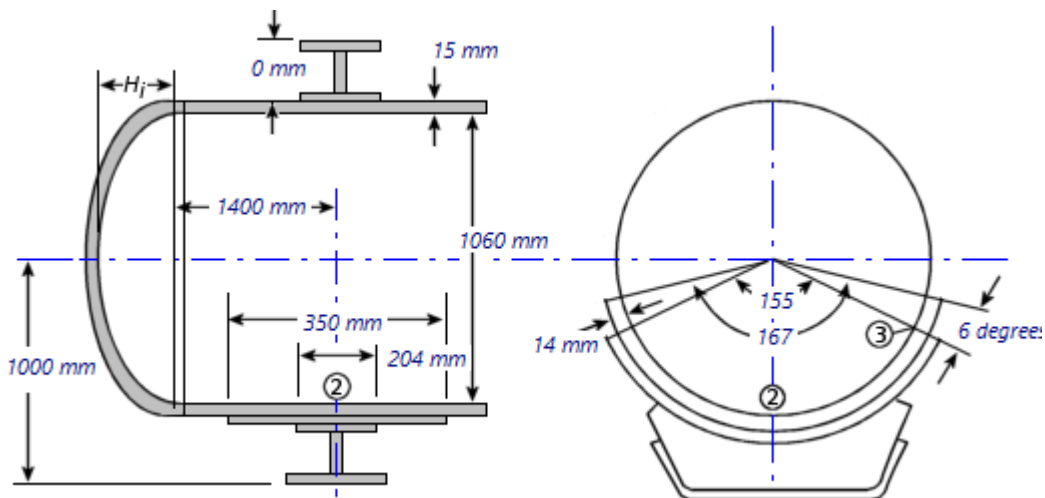
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	537.50	mm
Shell Thickness used in this Case	t	15.000	mm
Stiffened Vessel Length per 4.15.6	L	570.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1400.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	155.00	degrees
Wear Plate Width	b2 or b1	350.00	mm
Wear Plate Bearing Angle	delta2 or theta1	167.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		23415.63	kgf
Pressure used in Analysis	P	50.836	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		902.15	2386.01
Long. Stress at Bottom of Midspan		919.47	2386.01
Long. Stress at Top of Saddles		1231.16	2386.01
Long. Stress at Bottom of Saddles		725.20	2386.01
Tangential Shear in Shell		111.46	1413.05
Circ. Stress at Horn of Saddle		179.21	3579.01
Circ. Compressive Stress in Shell		15.56	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (305/2 + 0) \cdot 1000/1091$$

$$= 418.7 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(78.9, 0) \cdot 1000/2900$$

$$= 27.2 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 22997 + \max(27.2, 419, 0, 0)$$

$$= 23415.6 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4 (1.09 \cdot 1.2)^2)$$

$$= 2345.980 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	23676.95	kgf
Transverse Shear Load	152.34	kgf
Longitudinal Shear Load	78.94	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1060/2 + 15 + 14) \sin(155/2), 2 \cdot 538)$$

$$= 1091.499 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1710	K2 = 0.7543	K3 = 0.4386	K4 = 0.2805
K5 = 0.6630	K6 = 0.0289	K7 = 0.0289	K8 = 0.2942
K9 = 0.2072	K10 = 0.0316	K1* = 0.2951	K6p = 0.0229
K7p = 0.0229			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -23416 \cdot 140 [1 - (1 - 140/570 + (53.8^2 - 0^2) / (2 \cdot 140 \cdot 570)) / (1 + (4 \cdot 0) / (3 \cdot 570))]$$

$$= -7458.3 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 23416 \cdot 570 / 4 ((1 + 2(53.8^2 - 0^2)/(570^2)) / (1 + (4 \cdot 0) / (3 \cdot 570)) - 4 \cdot 140/570)$$

$$= 1178.8 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 50.8 \cdot 538 / (2 \cdot 15) - 1179 / (\pi \cdot 538^2 \cdot 15)$$

$$= 902.15 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 50.8 \cdot 538 / (2 \cdot 15) + 1179 / (\pi \cdot 538^2 \cdot 15)$$

$$= 919.47 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 50.8 \cdot 538 / (2 \cdot 15) - 7458 / (0.17 \cdot \pi \cdot 538^2 \cdot 15)$$

$$= 1231.16 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 50.8 \cdot 538 / (2 \cdot 15) + 7458 / (0.3 \cdot \pi \cdot 538^2 \cdot 15)$$

$$= 725.20 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 23416 (570 - 2 \cdot 140) / (570 + (4 \cdot 0 / 3))$$

$$= 11913.2 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.75 \cdot 11913 / (538 \cdot 15)$$

$$= 111.46 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(538 \cdot 15)}$$

$$= 70.037 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(538 \cdot 15)}, 2 \cdot 140)$$

$$= 344.07 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.66 \cdot 23416 \cdot 0.1 / (344(15 + 1 \cdot 14))$$

$$= -15.56 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -23416 / (4(15 + 1 \cdot 14)344) -$$

$$3 \cdot 0.029 \cdot 23416 / (2(15 + 1 \cdot 14)^2)$$

$$= -179.21 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2243.3	kgf/cm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(530 \cdot 15)} + 350)15^3) / 12$$

$$= 13.756 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 13.8 + 73.4(182 - 7.5)^2$$

$$= 22411.445 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	489.1	15.0	7.5	73.4	55023.1	0.224E+05
Wearplate	350.0	14.0	22.0	49.0	107800.0	0.126E+05
Web	14.0	416.0	237.0	58.2	1380288.0	0.101E+05
BasePlate	254.0	25.0	457.5	63.5	2905125.0	0.482E+05
Totals	...	...	...	244.1	4448236.0	0.933E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4448236/244$$

$$= 182.227 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 155/2$$

$$= 102.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(102) - 0.5 \cdot \sin(102)^2) / (\pi - 1.79 + \sin(102) \cos(102))$$

$$= 0.2690$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.27 \cdot 23416$$

$$= 6298.1499 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 36.8874 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1345.9545 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 530 \cdot \sin(155/2) / 1.35$$

$$= 617.459 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3888.8521 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 75.9579 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1495.5050 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(23416 + 261) \cdot 254) / (4 \cdot 1200 \cdot 1496))}$$

$$= 15.854 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 155/2) (530 + 15 + 14)$$

$$= 1091.499 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1091 / (3 - 1)$$

$$= 545.749 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 546 \cdot 254/2$$

$$= 693.102 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 23677 / (1200 \cdot 254)$$

$$= 7.768 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 693 \cdot 7.77$$

$$= 5384.034 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 26.6 + 38.2$$

$$= 64.802 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 5384/64.8$$

$$= 83.084 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	204.0	...	28.6	...	990.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 14$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/64.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(990/64.8)}$$

$$= 39.095 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(155/2) (530 + 15 + 14) - 25$$

$$= 854.010 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/2243)}$$

$$= 133.944$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 854/39.1$$

$$= 21.844$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 78.9 / (2 \cdot 1200) \cdot 546 \cdot 854/2$$

$$= 7.665 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($21.8 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (21.8)^2 / (2 \cdot 134^2)) 2243 /$$

$$(5/3 + 3 \cdot (21.8) / (8 \cdot 134) - (21.8)^3 / (8 \cdot 134^3))$$

$$= 1281 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 83.1/1281 + (7.67 \cdot 102/9904609) / 1496$$

$$= 0.070$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	190.0	0.0	26.6	0.0	990.



Web	545.7	14.0	0.0	76.4	0.0	12.5
Totals	...	...	...	103.0	...	0.100E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/103$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(559/2)(25 + 0 + 0) - 0$$

$$= 416.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1003/103)}$$

$$= 31.199 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 416/31.2$$

$$= 13.334$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 78.9 / (2 \cdot 1200)$$

$$= 0.033 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.033 \cdot 546 \cdot 416$$

$$= 7.468 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 7.47/98296$$

$$= 7.597 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($13.3 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.3)^2 / (2 \cdot 134^2)) 2243 /$$

$$(5/3 + 3 \cdot (13.3) / (8 \cdot 134) - (13.3^3) / (8 \cdot 134^3))$$

$$= 1310 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 101/1310 + (7.47 \cdot 102/1003) / 1496$$

$$= 0.082$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	1203.20 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	23258.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	27.2 kgf
Maximum Transverse Force	Ft	152.3 kgf
Maximum Longitudinal Force	Fl	78.9 kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $78.9 / (4.87 \cdot 4)$
 = 4.1 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $152 / (4.87 \cdot 4)$
 = 7.8 kgf/cm². Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $78.9 / (1052 \cdot 4)$
 = 0.0188 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1000 \cdot 152 + 0$
 = 152.34 kgf-m.

Eccentricity (e):
 = R_{mom} / QO
 = $152 / 23258$
 = 6.55 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.019, 0]
 = 0.0188 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	537.50	mm
Shell Thickness used in this Case	t	15.000	mm
Stiffened Vessel Length per 4.15.6	L	570.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1400.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	155.00	degrees
Wear Plate Width	b2 or b1	350.00	mm
Wear Plate Bearing Angle	delta2 or theta1	167.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	27.00	cm
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 7667.88 kgf
 Pressure used in Analysis P 50.836 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	913.17	2386.01
Long. Stress at Bottom of Midspan	908.46	2386.01
Long. Stress at Top of Saddles	1038.86	2386.01
Long. Stress at Bottom of Saddles	836.62	2386.01
Tangential Shear in Shell	34.33	1413.05
Circ. Stress at Horn of Saddle	58.69	3579.01
Circ. Compressive Stress in Shell	5.09	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (305/2 + 0) \cdot 1000/1091$
 $= 418.7 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(78.9, 0) \cdot 1000/2900$
 $= 27.2 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 7249 + \max(27.2, 419, 0, 0)$
 $= 7667.9 \text{ kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4 (1.09 \cdot 1.2)^2)$
 $= 2345.980 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7929.20	kgf
Transverse Shear Load	152.34	kgf
Longitudinal Shear Load	78.94	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1060/2 + 15 + 14) \sin(155/2), 2 \cdot 538)$
 $= 1091.499 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1710	K2 = 0.7543	K3 = 0.4386	K4 = 0.2805
K5 = 0.6630	K6 = 0.0289	K7 = 0.0289	K8 = 0.2942
K9 = 0.2072	K10 = 0.0316	K1* = 0.2951	K6p = 0.0229
K7p = 0.0229			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -7668 \cdot 140 [1 - (1 - 140/570 + (53.8^2 - 27^2) / (2 \cdot 140 \cdot 570)) / (1 + (4 \cdot 27)/(3 \cdot 570))]$
 $= -2981.1 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 7668 \cdot 570 / 4 ((1 + 2(53.8^2 - 27^2)/(570^2)) / (1 + (4 \cdot 27)/$



$$(3 \cdot 570) - 4 \cdot 140 / 570 \\ = -320.8 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 50.8 \cdot 538 / (2 \cdot 15) - 321 / (\pi \cdot 538^2 \cdot 15) \\ = 913.17 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 50.8 \cdot 538 / (2 \cdot 15) + 321 / (\pi \cdot 538^2 \cdot 15) \\ = 908.46 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 50.8 \cdot 538 / (2 \cdot 15) - 2981 / (0.17 \cdot \pi \cdot 538^2 \cdot 15) \\ = 1038.86 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 50.8 \cdot 538 / (2 \cdot 15) + 2981 / (0.3 \cdot \pi \cdot 538^2 \cdot 15) \\ = 836.62 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 7668 (570 - 2 \cdot 140) / (570 + (4 \cdot 27 / 3)) \\ = 3669.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 0.75 \cdot 3669 / (538 \cdot 15) \\ = 34.33 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)} \\ = 0.78 \cdot \sqrt{(538 \cdot 15)} \\ = 70.037 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\ = \min(204 + 1.56 \cdot \sqrt{(538 \cdot 15)}, 2 \cdot 140) \\ = 344.07 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.66 \cdot 7668 \cdot 0.1 / (344(15 + 1 \cdot 14)) \\ = -5.09 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -7668 / (4(15 + 1 \cdot 14)344) - 3 \cdot 0.029 \cdot 7668 / (2(15 + 1 \cdot 14)^2) \\ = -58.69 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2243.3	kgf/cm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(530 \cdot 15)} + 350)^3) / 12$
 $= 13.756 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 13.8 + 73.4 (182 - 7.5)^2$
 $= 22411.445 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	489.1	15.0	7.5	73.4	55023.1	0.224E+05
Wearplate	350.0	14.0	22.0	49.0	107800.0	0.126E+05
Web	14.0	416.0	237.0	58.2	1380288.0	0.101E+05
BasePlate	254.0	25.0	457.5	63.5	2905125.0	0.482E+05
Totals	...	...	...	244.1	4448236.0	0.933E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 4448236 / 244$
 $= 182.227 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 155 / 2$
 $= 102.5$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(102) - 0.5 \cdot \sin(102)^2) / (\pi - 1.79 + \sin(102) \cos(102))$
 $= 0.2690$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.27 \cdot 7668$
 $= 2062.4451 \text{ kgf}$

Tension Stress, St = (Fh/As) = 12.0795 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1345.9545 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1000 - 530 \cdot \sin(155/2) / 1.35$
 $= 617.459 \text{ mm}$

Bending Moment, M = Fh * d = 1273.4762 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 24.8738 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1495.5050 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(7668 + 261)254) / (4 \cdot 1200 \cdot 1496)}$
 $= 9.174 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 155/2) (530 + 15 + 14)$
 $= 1091.499 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1091 / (3 - 1)$
 $= 545.749 \text{ mm}$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 546 \cdot 254 / 2$$

$$= 693.102 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 7929 / (1200 \cdot 254)$$

$$= 2.601 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 693 \cdot 2.6$$

$$= 1803.065 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 26.6 + 38.2$$

$$= 64.802 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 1803 / 64.8$$

$$= 27.824 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	204.0	...	28.6	...	990.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 14$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 64.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204 / 2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (990 / 64.8) }$$

$$= 39.095 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(155 / 2) (530 + 15 + 14) - 25$$

$$= 854.010 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 2038900 / 2243) }$$

$$= 133.944$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 854 / 39.1$$

$$= 21.844$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 78.9 / (2 \cdot 1200) \cdot 546 \cdot 854 / 2$$

$$= 7.665 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (21.8 < 134) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (21.8)^2 / (2 \cdot 134^2)) 2243 / (5/3 + 3 \cdot (21.8) / (8 \cdot 134) - (21.8^3) / (8 \cdot 134^3))$$

$$= 1281 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 27.8 / 1281 + (7.67 \cdot 102 / 9904609) / 1496$$

$$= 0.027$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	190.0	0.0	26.6	0.0	990.
Web	545.7	14.0	0.0	76.4	0.0	12.5
Totals	...	...	...	103.0	...	0.100E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 103$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204 / 2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000 - \cos(559/2) (25 + 0 + 0) - 0$$

$$= 416.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1003 / 103) }$$

$$= 31.199 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 416 / 31.2$$

$$= 13.334$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 78.9 / (2 \cdot 1200)$$

$$= 0.033 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.033 \cdot 546 \cdot 416$$

$$= 7.468 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 7.47 / 98296$$

$$= 7.597 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (13.3 < 134) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.3)^2 / (2 \cdot 134^2)) 2243 / (5/3 + 3 \cdot (13.3) / (8 \cdot 134) - (13.3^3) / (8 \cdot 134^3))$$

$$= 1310 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 33.9/1310 + (7.47 \cdot 102/1003)/1496$$

$$= 0.031$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	1203.20 kgf/cm ²
Area Available in a Single Bolt	BlArea	4.8662 cm ²
Saddle Load QO (Weight)	QO	7510.5 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	27.2 kgf
Maximum Transverse Force	Ft	152.3 kgf
Maximum Longitudinal Force	Fl	78.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 78.9 / (4.87 \cdot 4)$$

$$= 4.1 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 152 / (4.87 \cdot 4)$$

$$= 7.8 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 78.9 / (1052 \cdot 4)$$

$$= 0.0188 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 152 + 0$$

$$= 152.34 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 152/7510$$

$$= 20.28 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.019, 0]$$

$$= 0.0188 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: T3-Tube Inlet

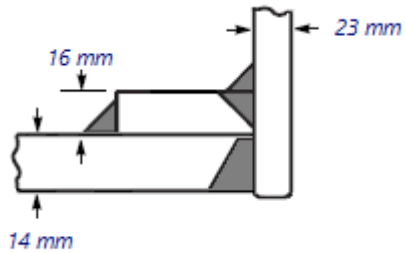
From: 30

Pressure for Reinforcement Calculations	P	20.501	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		58.12	cm
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 60	
Pad Allowable Stress at Temperature	Sp	1203.20	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1203.20	kgf/cm ²
Diameter of Pad along vessel surface	Dp	319.0750	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: T3-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	173.050 mm.
Actual Thickness Used in Calculation	23.012 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 20.5 \cdot 533 / (1407 \cdot 1 - 0.6 \cdot 20.5)$
 $= 7.8339 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 20.5 \cdot 89.5 / (1203 \cdot 1 - 0.6 \cdot 20.5)$
 $= 1.5411 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	358.1004 mm
Parallel to Vessel Wall, opening length	d	179.0502 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 0.86)$
 $= 0.855$

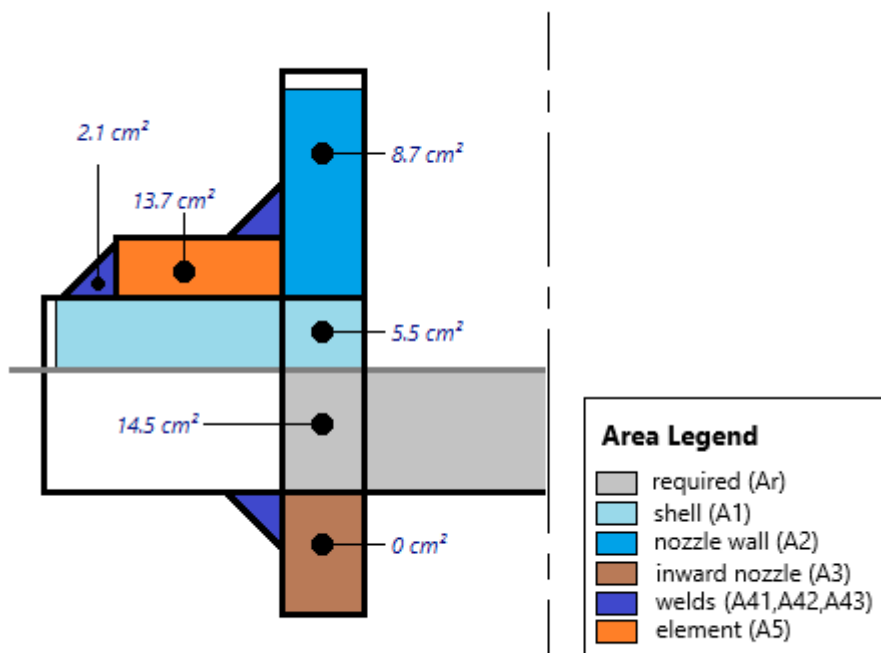


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.481	NA	NA
Area in Shell	A1	5.485	NA	NA
Area in Nozzle Wall	A2	8.687	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.086	NA	NA
Area in Element	A5	13.681	NA	NA
TOTAL AREA AVAILABLE	Atot	29.940	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (179 \cdot 7.83 \cdot 1 + 2 \cdot 20 \cdot 7.83 \cdot 1 \cdot (1 - 0.86)) \\
 &= 14.481 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 179(1 \cdot 11 - 1 \cdot 7.83) - 2 \cdot 20 \\
 &\quad (1 \cdot 11 - 1 \cdot 7.83) \cdot (1 - 0.86) \\
 &= 5.485 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 27.5) \cdot (20 - 1.54) \cdot 0.86 \\
 &= 8.687 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 0.86
 \end{aligned}$$

= 2.086 cm²

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_r$$

$$= (319 - 219) \cdot 16 \cdot 0.86$$

$$= 13.681 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5411 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.8339 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.8339 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.2, \max(10.8, 5.5)]$$

$$= 10.1600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.54, 10.2)$$

$$= 10.1600 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 23.012 = 20.136 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.3	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0	Allowable	: 2978.7 kgf/cm ²	Passed
Occasional	: 41.2	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 34.0	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 20.1, tr = 1.54, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.09$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: C

Govrn. thk, tg = 16, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.71$, Temp. Reduction = 16 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve C	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 14, tr = 7.83, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.71$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-36 °C
--	--------

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle : -46 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 20.50/52.01 = 0.394

Weld Size Calculations, Description: T3-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	5.5000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (14.5 - 5.49 + 2 \cdot 20 \cdot 0.86 \cdot (1 \cdot 11 - 7.83)) \cdot 1407)$
 $= 14182.77 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (8.69 + 13.7 + 2.09 - 0 \cdot 0.86) \cdot 1407$
 $= 34410.56 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (8.69 + 0 + 0.86 + (3.76)) \cdot 1407$
 $= 18724.11 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (8.69 + 0 + 2.09 + 13.7 + (3.76)) \cdot 1407$
 $= 39707.93 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 1203$
 $= 20288. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 319 \cdot 12 \cdot 0.49 \cdot 1203$
 $= 35459. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$



$$= (3.14 \cdot 99.5) \cdot (23 - 3) \cdot 0.7 \cdot 1203$$

$$= 52704. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 219 \cdot 16 \cdot 0.74 \cdot 1203$$

$$= 49023. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 219 \cdot (14 - 3) \cdot 0.74 \cdot 1407$$

$$= 39416. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (35459 + 52704) = 88163 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (20288 + 49023 + 39416 + 0) = 108727 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (35459 + 39416 + 0) = 74875 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 88163 kgf, must exceed W = 14182 kgf or W1 = 34410 kgf
 Path 2-2 = 108727 kgf, must exceed W = 14182 kgf or W2 = 18724 kgf
 Path 3-3 = 74874 kgf, must exceed W = 14182 kgf or W3 = 39707 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28.6 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.4428 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.4428 mm

Input Echo, WRC297 Item 1, Description: T3-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1060.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	173.050	mm
Nozzle Thickness	23.0124	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	319.075	mm
Thickness of Reinforcing Pad	16.0000	mm



Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	20.50	kgf/cm ²
Radial Load	P	-1529.65	kgf
Circumferential Shear	Vc	1529.65	kgf
Longitudinal Shear	Vl	1147.24	kgf
Circumferential Moment	Mc	0.99	kgf-m.
Longitudinal Moment	Ml	0.99	kgf-m.
Torsional Moment	Mt	0.11	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.520) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(546.5(14.0 - 3.0))}$$

$$= 474.937 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1093.000 mm
Nozzle Outside Diameter	(d) =	219.075 mm
Vessel Thickness used	(T) =	27.000 mm
Nozzle Thickness used	(t) =	20.012 mm
T / t	=	1.349
d / t	=	10.947

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.275$$

Nr/P	=	0.118
Mr/P	=	0.147
M0/P	=	0.058
N0/P	=	0.129
MrD/Mc	=	0.250
NrDL/Mc	=	0.158
M0d/Mc	=	0.118
N0DL/Mc	=	0.050
MrD/Ml	=	0.166
NrDL/Ml	=	0.114
M0D/Ml	=	0.080
N0DL/Ml	=	0.147

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	26	26	26	26
Outplane Bending (P)	73	-73	73	-73
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	394	415	394	415
Outplane Stress Summary	494	368	495	368

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	24	24	24	24
Inplane Bending (P)	184	-184	184	-184
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	197	197	197	197

Inplane Stress Summary	406	37	407	36
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	16	16	-16	-16
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	16	16	-16	-16
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	497	369	498	369
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	26	26	26	26
Outplane Bending (P)	73	-73	73	-73
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	197	197	197	197

Outplane Stress Summary	297	151	298	150
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	24	24	24	24
Inplane Bending (P)	184	-184	184	-184
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	394	415	394	415

Inplane Stress Summary	603	255	605	254
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-12	-12	12	12
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	603	257	605	255

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	26	26	26	26
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	98	123	98	123
Hoop Stress Summary	125	150	125	150

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	12	12	12	12
Axial Bending (P)	201	-201	201	-201
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	49	49	49	49
Axial Stress Summary	262	-139	263	-140

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	22	22	-22	-22
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	22	22	-22	-22

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	265	292	267	294

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	26	26	26	26
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	98	123	98	123
Hoop Stress Summary	125	150	125	150

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	12	12	12	12
Axial Bending (P)	201	-201	201	-201
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	49	49	49	49
Axial Stress Summary	261	-138	264	-140

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-16	-16	16	16
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-16	-16	16	16

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	263	291	266	293

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(538.5(14.0 - 3.0))}$
 $= 473.057 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	1077.000 mm
Nozzle Outside Diameter (d)	=	319.075 mm
Vessel Thickness used (T)	=	11.000 mm
Nozzle Thickness used (t)	=	70.012 mm
T / t	=	0.157
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.931$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.109
M0/P	=	0.032
N0/P	=	0.054
MrD/Mc	=	0.269
NrDL/Mc	=	0.100
M0d/Mc	=	0.081
N0DL/Mc	=	0.076
MrD/Ml	=	0.083
NrDL/Ml	=	0.074
M0D/Ml	=	0.024
N0DL/Ml	=	0.068

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	68	68	68	68
Outplane Bending (P)	244	-244	244	-244



Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	983	1003	983	1003

Outplane Stress Summary	1295	828	1296	827
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	77	77	77	77
Inplane Bending (P)	825	-825	825	-825
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	491	491	491	491

Inplane Stress Summary	1393	-255	1395	-257
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	27	27	-27	-27
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	27	27	-27	-27
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1400	1084	1403	1086
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	68	68	68	68
Outplane Bending (P)	244	-244	244	-244
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-1	1	1	-1
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	491	491	491	491

Outplane Stress Summary	802	316	805	314
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	77	77	77	77
Inplane Bending (P)	825	-825	825	-825
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	983	1003	983	1003

Inplane Stress Summary	1881	259	1890	251
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CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-20	-20	20	20
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			-20	-20	20	20

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1882	323	1890	320

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	394	415	394	415	394	415	394	415
Circ. Pl (SUS)	26	26	26	26	24	24	24	24
Circ. Q (SUS)	73	-73	73	-73	184	-184	184	-184
Long. Pm (SUS)	197	197	197	197	197	197	197	197
Long. Pl (SUS)	24	24	24	24	26	26	26	26
Long. Q (SUS)	184	-184	184	-184	73	-73	73	-73
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	16	16	-16	-16	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	394.0	415.0	394.0	415.0	394.0	415.0	394.0	415.0
Pm+Pl (SUS)	421.3	442.2	421.3	442.2	418.7	439.7	418.7	439.7
Pm+Pl+Q (Total)	495.8	368.8	495.8	368.8	602.5	256.4	602.5	256.4

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	415.00	1407.13	Passed
Pm+Pl (SUS)	442.16	2110.69	Passed
Pm+Pl+Q (TOTAL)	602.47	4221.39	Passed

*Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.*

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

[illegible]



Long. P1 (SUS)	77	77	77	77	68	68	68	68
Long. Q (SUS)	824	-824	826	-826	243	-243	245	-245
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	27	27	-27	-27	-20	-20	20	20
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	983	1003	983	1003	983	1003	983	1003
Pm+P1 (SUS)	1052	1072	1052	1072	1060	1080	1060	1080
Pm+P1+Q (Total)	1399	1084	1400	1086	1881	322	1889	319

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1003.00	1407.13	Passed
Pm+P1 (SUS)	1080.77	2110.69	Passed
Pm+P1+Q (TOTAL)	1889.37	4221.39	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	98	123	98	123	98	123	98	123
Circ. P1 (SUS)	26	26	26	26	26	26	26	26
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	49	49	49	49	49	49	49	49
Long. P1 (SUS)	12	12	12	12	12	12	12	12
Long. Q (SUS)	201	-201	201	-201	200	-200	202	-202
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	22	22	-22	-22	-16	-16	16	16
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	98.0	123.0	98.0	123.0	98.0	123.0	98.0	123.0
Pm+P1 (SUS)	130.9	154.2	130.9	154.2	127.8	151.8	127.8	151.8
Pm+P1+Q (Total)	265.4	292.3	265.4	292.3	262.8	289.8	264.8	291.8

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.00	1203.20	Passed
Pm+P1 (SUS)	154.19	1804.80	Passed
Pm+P1+Q (TOTAL)	292.33	3609.61	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: T4-Tube outlet

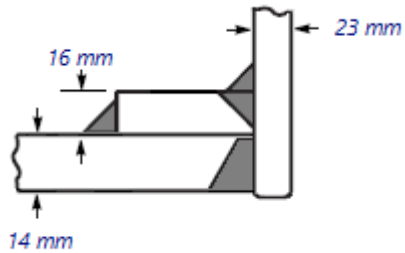
From: 30

Pressure for Reinforcement Calculations	P	20.395	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		58.12	cm
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 60	
Pad Allowable Stress at Temperature	Sp	1203.20	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1203.20	kgf/cm ²
Diameter of Pad along vessel surface	Dp	319.0750	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: T4-Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	173.050 mm.
Actual Thickness Used in Calculation	23.012 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 20.4 \cdot 533 / (1407 \cdot 1 - 0.6 \cdot 20.4)$
 $= 7.7930 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 20.4 \cdot 89.5 / (1203 \cdot 1 - 0.6 \cdot 20.4)$
 $= 1.5331 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	358.1004 mm
Parallel to Vessel Wall, opening length	d	179.0502 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 0.86)$
 $= 0.855$

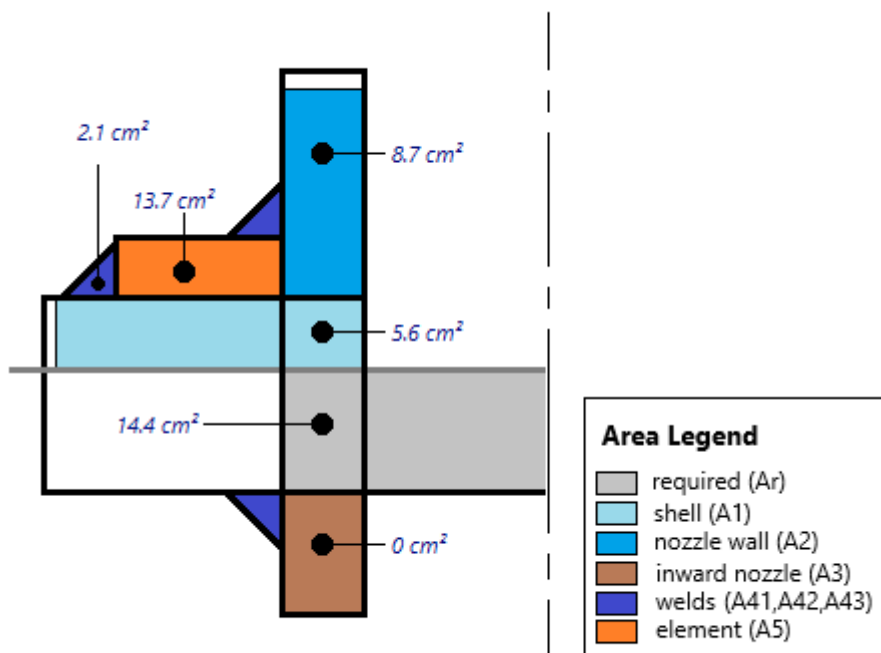


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.405	NA	NA
Area in Shell	A1	5.556	NA	NA
Area in Nozzle Wall	A2	8.691	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.086	NA	NA
Area in Element	A5	13.681	NA	NA
TOTAL AREA AVAILABLE	Atot	30.014	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (179 \cdot 7.79 \cdot 1 + 2 \cdot 20 \cdot 7.79 \cdot 1 \cdot (1 - 0.86)) \\
 &= 14.405 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 179(1 \cdot 11 - 1 \cdot 7.79) - 2 \cdot 20 \\
 &\quad (1 \cdot 11 - 1 \cdot 7.79) \cdot (1 - 0.86) \\
 &= 5.556 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 27.5) \cdot (20 - 1.53) \cdot 0.86 \\
 &= 8.691 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 0.86
 \end{aligned}$$

= 2.086 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
 = (319 - 219) • 16 • 0.86
 = 13.681 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 4.5331 mm
 Wall Thickness per UG16(b), tr16b = 5.5000 mm
 Wall Thickness, shell/head, internal pressure trb1 = 10.7930 mm
 Wall Thickness tb1 = max(trb1, tr16b) = 10.7930 mm
 Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm
 Wall Thickness per table UG-45 tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[10.2, max(10.8, 5.5)]
 = 10.1600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(4.53, 10.2)
 = 10.1600 mm

Available Nozzle Neck Thickness = 0.875 * 23.012 = 20.136 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	29.1,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2978.9 kgf/cm ²	Passed
Occasional	:	41.0,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	34.1,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govern. thk, tg = 20.1, tr = 1.53, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.089, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: C

Govern. thk, tg = 16, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.71, Temp. Reduction = 16 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve C	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: C

Govern. thk, tg = 14, tr = 7.79, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.71, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-36 °C
--	--------

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle : -46 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 20.39/52.01 = 0.392

Weld Size Calculations, Description: T4-Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	5.5000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (14.4 - 5.56 + 2 \cdot 20 \cdot 0.86 \cdot (1 \cdot 11 - 7.79)) \cdot 1407)$
 $= 13996.64 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (8.69 + 13.7 + 2.09 - 0 \cdot 0.86) \cdot 1407$
 $= 34415.89 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (8.69 + 0 + 0.86 + (3.76)) \cdot 1407$
 $= 18729.44 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (8.69 + 0 + 2.09 + 13.7 + (3.76)) \cdot 1407$
 $= 39713.25 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 1203$
 $= 20288. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 319 \cdot 12 \cdot 0.49 \cdot 1203$
 $= 35459. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$

$$= (3.14 \cdot 99.5) \cdot (23 - 3) \cdot 0.7 \cdot 1203$$

$$= 52704. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 219 \cdot 16 \cdot 0.74 \cdot 1203$$

$$= 49023. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 219 \cdot (14 - 3) \cdot 0.74 \cdot 1407$$

$$= 39416. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (35459 + 52704) = 88163 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (20288 + 49023 + 39416 + 0) = 108727 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (35459 + 39416 + 0) = 74875 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 88163 kgf, must exceed W = 13996 kgf or W1 = 34415 kgf
 Path 2-2 = 108727 kgf, must exceed W = 13996 kgf or W2 = 18729 kgf
 Path 3-3 = 74874 kgf, must exceed W = 13996 kgf or W3 = 39713 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 28.6 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.4428 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.4428 mm

Input Echo, WRC297 Item 1, Description: T4-Tube outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1060.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	173.050	mm
Nozzle Thickness	23.0124	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	319.075	mm
Thickness of Reinforcing Pad	16.0000	mm



Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	20.39	kgf/cm ²
Radial Load	P	-1529.65	kgf
Circumferential Shear	Vc	1529.65	kgf
Longitudinal Shear	Vl	1147.24	kgf
Circumferential Moment	Mc	0.99	kgf-m.
Longitudinal Moment	Ml	0.99	kgf-m.
Torsional Moment	Mt	1.15	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.520) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(546.5(14.0 - 3.0))}$$

$$= 474.937 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1093.000 mm
Nozzle Outside Diameter	(d) =	219.075 mm
Vessel Thickness used	(T) =	27.000 mm
Nozzle Thickness used	(t) =	20.012 mm
T / t	=	1.349
d / t	=	10.947

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.275$$

Nr/P	=	0.118
Mr/P	=	0.147
M0/P	=	0.058
N0/P	=	0.129
MrD/Mc	=	0.250
NrDL/Mc	=	0.158
M0d/Mc	=	0.118
N0DL/Mc	=	0.050
MrD/Ml	=	0.166
NrDL/Ml	=	0.114
M0D/Ml	=	0.080
N0DL/Ml	=	0.147

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	26	26	26	26
Outplane Bending (P)	73	-73	73	-73
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	392	413	392	413
Outplane Stress Summary	492	366	493	366

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	24	24	24	24
Inplane Bending (P)	184	-184	184	-184
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	196	196	196	196

Inplane Stress Summary	405	36	406	35
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	16	16	-16	-16
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	16	16	-16	-16
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	495	367	496	367
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	26	26	26	26
Outplane Bending (P)	73	-73	73	-73
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	196	196	196	196

Outplane Stress Summary	296	150	297	149
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	24	24	24	24
Inplane Bending (P)	184	-184	184	-184
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	392	413	392	413

Inplane Stress Summary	601	253	603	252
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-12	-12	12	12
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	601	255	603	253

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	26	26	26	26
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	98	122	98	122
Hoop Stress Summary	125	149	125	149

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	12	12	12	12
Axial Bending (P)	201	-201	201	-201
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	49	49	49	49
Axial Stress Summary	261	-139	263	-140

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	22	22	-22	-22
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	22	22	-22	-22

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	265	292	266	293

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	26	26	26	26
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	98	122	98	122
Hoop Stress Summary	125	149	125	149

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	12	12	12	12
Axial Bending (P)	201	-201	201	-201
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	49	49	49	49
Axial Stress Summary	261	-138	264	-140

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-16	-16	16	16
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-16	-16	16	16

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	263	290	265	292

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(538.5(14.0 - 3.0))}$
 $= 473.057 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) =	1077.000 mm
Nozzle Outside Diameter (d) =	319.075 mm
Vessel Thickness used (T) =	11.000 mm
Nozzle Thickness used (t) =	70.012 mm
T / t	= 0.157
d / t	= 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.931$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.109
M0/P	=	0.032
N0/P	=	0.054
MrD/Mc	=	0.269
NrDL/Mc	=	0.100
M0d/Mc	=	0.081
N0DL/Mc	=	0.076
MrD/Ml	=	0.083
NrDL/Ml	=	0.074
M0D/Ml	=	0.024
N0DL/Ml	=	0.068

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	68	68	68	68
Outplane Bending (P)	244	-244	244	-244



Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	978	998	978	998

Outplane Stress Summary	1290	822	1291	822
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	77	77	77	77
Inplane Bending (P)	825	-825	825	-825
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	489	489	489	489

Inplane Stress Summary	1390	-258	1393	-260
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	27	27	-27	-27
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	27	27	-27	-27
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1397	1082	1400	1084
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	68	68	68	68
Outplane Bending (P)	244	-244	244	-244
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-1	1	1	-1
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	489	489	489	489

Outplane Stress Summary	800	314	802	312
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	77	77	77	77
Inplane Bending (P)	825	-825	825	-825
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	978	998	978	998

Inplane Stress Summary	1876	254	1885	246
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CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-20	-20	20	20
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			-20	-20	20	20

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1877	320	1885	318

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	392	413	392	413	392	413	392	413
Circ. Pl (SUS)	26	26	26	26	24	24	24	24
Circ. Q (SUS)	73	-73	73	-73	184	-184	184	-184
Long. Pm (SUS)	196	196	196	196	196	196	196	196
Long. Pl (SUS)	24	24	24	24	26	26	26	26
Long. Q (SUS)	184	-184	184	-184	73	-73	73	-73
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	16	16	-16	-16	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	392.0	413.0	392.0	413.0	392.0	413.0	392.0	413.0
Pm+Pl (SUS)	419.3	440.2	419.3	440.2	416.7	437.7	416.7	437.7
Pm+Pl+Q (Total)	493.8	366.8	493.8	366.8	600.5	254.4	600.5	254.4

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	413.00	1407.13	Passed
Pm+Pl (SUS)	440.16	2110.69	Passed
Pm+Pl+Q (TOTAL)	600.47	4221.39	Passed

*Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.*

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	978	998	978	998	978	998	978	998
Circ. Pl (SUS)	68	68	68	68	77	77	77	77
Circ. Q (SUS)	244	-244	244	-244	821	-821	829	-829
Long. Pm (SUS)	489	489	489	489	489	489	489	489

Long. P1 (SUS)	77	77	77	77	68	68	68	68
Long. Q (SUS)	824	-824	826	-826	243	-243	245	-245
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	27	27	-27	-27	-20	-20	20	20
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	978.0	998.0	978.0	998.0	978.0	998.0	978.0	998.0
Pm+P1 (SUS)	1047	1067	1047	1067	1055	1075	1055	1075
Pm+P1+Q (Total)	1396	1081	1398	1083	1876	320	1884	317

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	998.00	1407.13	Passed
Pm+P1 (SUS)	1075.77	2110.69	Passed
Pm+P1+Q (TOTAL)	1884.37	4221.39	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	98	122	98	122	98	122	98	122
Circ. P1 (SUS)	26	26	26	26	26	26	26	26
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	49	49	49	49	49	49	49	49
Long. P1 (SUS)	12	12	12	12	12	12	12	12
Long. Q (SUS)	201	-201	201	-201	200	-200	202	-202
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	22	22	-22	-22	-16	-16	16	16
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	98.0	122.0	98.0	122.0	98.0	122.0	98.0	122.0
Pm+P1 (SUS)	130.9	153.2	130.9	153.2	127.8	150.8	127.8	150.8
Pm+P1+Q (Total)	265.4	291.3	265.4	291.3	262.8	288.8	264.8	290.8

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	122.00	1203.20	Passed
Pm+P1 (SUS)	153.25	1804.80	Passed
Pm+P1+Q (TOTAL)	291.34	3609.61	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

Input, Nozzle Desc: S1-Shell Inlet

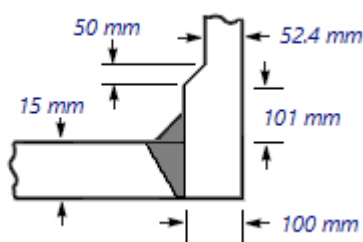
From: 40

Pressure for Reinforcement Calculations	P	25.494	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Shell Finished (Minimum) Thickness	t	15.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		605.12	cm
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		140	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	101.0000	mm
Height of Beveled Transition	L'	50.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: S1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 504.850 mm.
Actual Thickness Used in Calculation 52.375 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.5 \cdot 533 / (1407 \cdot 1 - 0.6 \cdot 25.5)$
= 9.7627 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.5 \cdot 255 / (1407 \cdot 1 - 0.6 \cdot 25.5)$
= 4.6785 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 47.6/0.5773
= 82.4893 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

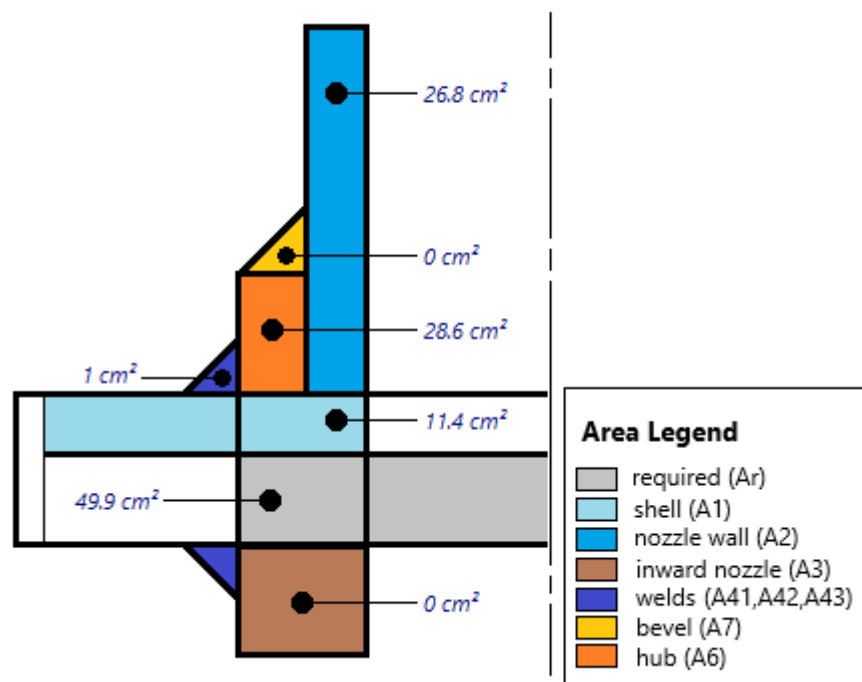
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1021.7008	mm
Parallel to Vessel Wall, opening length	d	510.8504	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1407/1407)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1407/1407)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	49.873	NA	NA
Area in Shell	A1	11.429	NA	NA
Area in Nozzle Wall	A2	26.818	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	28.575	NA	NA
TOTAL AREA AVAILABLE	Atot	67.822	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (511 \cdot 9.76 \cdot 1 + 2 \cdot 97 \cdot 9.76 \cdot 1 \cdot (1 - 1))$$

$$= 49.873 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 511 (1 \cdot 12 - 1 \cdot 9.76) - 2 \cdot 97 (1 \cdot 12 - 1 \cdot 9.76) \cdot (1 - 1)$$

$$= 11.429 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 30) (49.4 - 4.68) \cdot 1$$

$$= 26.818 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot \text{fr2}$$

$$= (2 \cdot \min(30, 300, 101)) \cdot (100 - 52.4) \cdot 1$$

$$= 28.575 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.6785 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.7627 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.7627 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(12.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.68, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 52.375 = 45.828 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 56.0,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3461.9 kgf/cm ²	Passed
Occasional	: 60.1,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 11.6,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 45.8, tr = 4.68, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	34 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $25.49/52.01 = 0.490$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 47.6/0.5773$$

= 82.4893 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 766.2756 mm
 Parallel to Vessel Wall, opening length d 383.1378 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 30.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	33.250	33.250
Area Available in Shell	A1	5.715	5.715
Area Available in Nozzle Wall	A2	26.818	26.818
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	28.575	28.575
TOTAL AREA AVAILABLE	Atot	62.108	62.108

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: S1-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (49.9 - 11.4 + 2 \cdot 97 \cdot 1 \cdot (1 \cdot 12 - 9.76)) \cdot 1407)$
 $= 60202.31 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (55.4 + 1 - 0 \cdot 1) \cdot 1407$
 $= 79351.98 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (55.4 + 0 + 1 + (23.3)) \cdot 1407$
 $= 112109.91 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (55.4 + 0 + 1 + 0 + (23.3)) \cdot 1407$
 $= 112109.91 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 705 \cdot 10 \cdot 0.49 \cdot 1407$
 $= 76339. \text{ kgf}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 304) \cdot (100 - 3) \cdot 0.7 \cdot 1407$$

$$= 912266. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 705 \cdot (15 - 3) \cdot 0.74 \cdot 1407$$

$$= 138345. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (76339 + 912266) = 988604 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (76339 + 0 + 138345 + 0) = 214684 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (76339 + 138345 + 0) = 214684 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 988604 kgf, must exceed W = 60202 kgf	or W1 = 79351 kgf
Path 2-2 = 214684 kgf, must exceed W = 60202 kgf	or W2 = 112109 kgf
Path 3-3 = 214684 kgf, must exceed W = 60202 kgf	or W3 = 112109 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 29.9 kgf/cm²

The Drop for this Nozzle is : 134.1509 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 449.1509 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 418.61 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 510.85 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: S1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1060.000	mm
Shell Thickness	15.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	504.850	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	1407.13 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.13 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	25.49	kgf/cm ²
Radial Load	P	-3671.16	kgf
Circumferential Shear	Vc	3671.16	kgf
Longitudinal Shear	Vl	2753.37	kgf
Circumferential Moment	Mc	5.72	kgf-m.
Longitudinal Moment	Ml	5.72	kgf-m.
Torsional Moment	Mt	6.61	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.647) is > 0.5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (7.049) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 704.85 + 2 \cdot 1.65 \cdot \sqrt{(539.0(15.0 - 3.0))}$
 $= 970.249 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1078.000 mm
Nozzle Outside Diameter	(d) =	704.850 mm
Vessel Thickness used	(T) =	12.000 mm
Nozzle Thickness used	(t) =	97.000 mm
T / t	=	0.124
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.197$$

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.034
MrD/Mc	=	0.241
NrDL/Mc	=	0.090
M0d/Mc	=	0.069
N0DL/Mc	=	0.061
MrD/Ml	=	0.045
NrDL/Ml	=	0.038
M0D/Ml	=	0.014
N0DL/Ml	=	0.030

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	87	87	87	87
Outplane Bending (P)	309	-309	309	-309
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	1119	1145	1119	1145
Outplane Stress Summary	1516	923	1517	922

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	111	111	111	111
Inplane Bending (P)	1065	-1065	1065	-1065
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	559	559	559	559
Inplane Stress Summary	1734	-393	1738	-395

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	27	27	-27	-27
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	27	27	-27	-27

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1738	1317	1741	1319

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	87	87	87	87
Outplane Bending (P)	309	-309	309	-309
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-2	2	2	-2
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	559	559	559	559
Outplane Stress Summary	954	339	959	335

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	111	111	111	111
Inplane Bending (P)	1065	-1065	1065	-1065
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-8	8	8	-8
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1119	1145	1119	1145
Inplane Stress Summary	2287	198	2304	183

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-20	-20	20	20
Torsional Shear (Mt)	0	0	0	0



Shear Stress Summary -20 -20 20 20

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2287	342	2305	338

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	87	87	87	87
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	67	98	67	98

Hoop Stress Summary 154 185 155 185

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	33	33	33	33

Axial Stress Summary 46 24 46 24

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 3 3 -3 -3

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	154	185	155	185

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	87	87	87	87
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	67	98	67	98

Hoop Stress Summary 154 185 155 185

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	33	33	33	33
Axial Stress Summary	46	24	47	24

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-2	-2	2	2

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	154	185	155	185

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:
Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1119	1145	1119	1145	1119	1145	1119	1145
Circ. Pl (SUS)	87	87	87	87	111	111	111	111
Circ. Q (SUS)	309	-309	309	-309	1057	-1057	1073	-1073
Long. Pm (SUS)	559	559	559	559	559	559	559	559
Long. Pl (SUS)	111	111	111	111	87	87	87	87
Long. Q (SUS)	1064	-1064	1066	-1066	307	-307	311	-311
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	27	27	-27	-27	-20	-20	20	20
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1119	1145	1119	1145	1119	1145	1119	1145
Pm+Pl (SUS)	1207	1233	1207	1233	1230	1256	1230	1256
Pm+Pl+Q (Total)	1737	1318	1739	1320	2287	341	2303	337

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1145.00	1407.13	Passed
Pm+Pl (SUS)	1256.66	2110.69	Passed
Pm+Pl+Q (TOTAL)	2303.30	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	67	98	67	98	67	98	67	98
Circ. Pl (SUS)	87	87	87	87	87	87	87	87
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	33	33	33	33	33	33	33	33
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	11	-11	11	-11	11	-11	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	67.0	98.0	67.0	98.0	67.0	98.0	67.0	98.0
Pm+Pl (SUS)	154.1	185.1	154.1	185.1	154.0	185.0	154.0	185.0
Pm+Pl+Q (Total)	154.1	185.1	154.1	185.1	154.0	185.0	154.0	185.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	98.00	1407.13	Passed
Pm+Pl (SUS)	185.06	2110.70	Passed
Pm+Pl+Q (TOTAL)	185.06	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: S2-Shell outlet

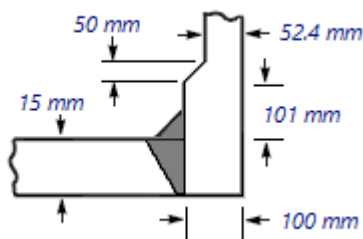
From: 40

Pressure for Reinforcement Calculations	P	25.494	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Shell Finished (Minimum) Thickness	t	15.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		175.12	cm
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		140	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	101.0000	mm
Height of Beveled Transition	L'	50.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: S2-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 504.850 mm.
 Actual Thickness Used in Calculation 52.375 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.5 \cdot 533 / (1407 \cdot 1 - 0.6 \cdot 25.5)$
 $= 9.7629$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.5 \cdot 255 / (1407 \cdot 1 - 0.6 \cdot 25.5)$
 $= 4.6786$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 47.6 / 0.5773$
 $= 82.4893$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

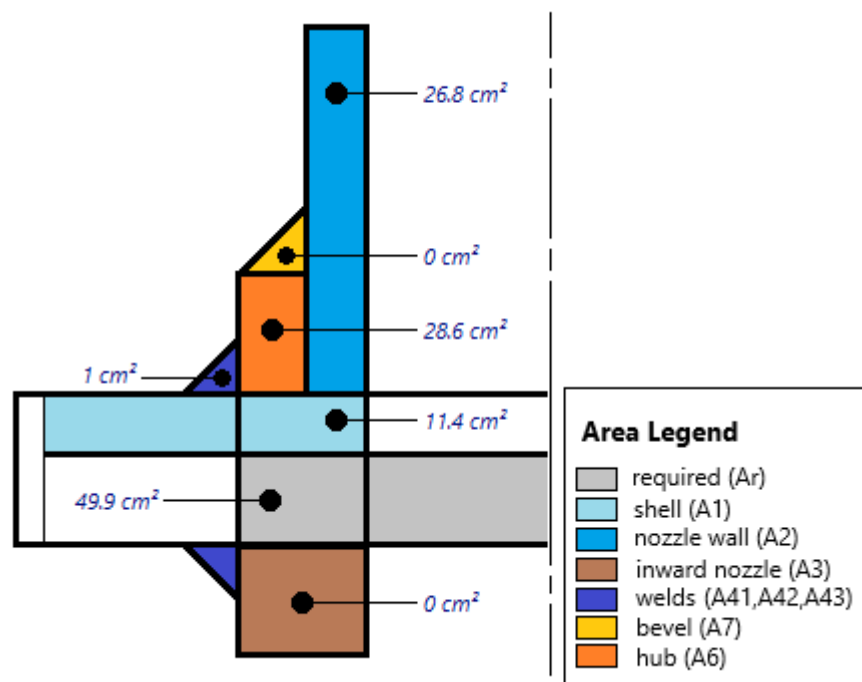
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1021.7008	mm
Parallel to Vessel Wall, opening length	d	510.8504	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1407)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1407)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	49.874	NA	NA
Area in Shell	A1	11.428	NA	NA
Area in Nozzle Wall	A2	26.818	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	28.575	NA	NA
TOTAL AREA AVAILABLE	Atot	67.821	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (511 \cdot 9.76 \cdot 1 + 2 \cdot 97 \cdot 9.76 \cdot 1 \cdot (1 - 1))$
 $= 49.874 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 511 (1 \cdot 12 - 1 \cdot 9.76) - 2 \cdot 97$
 $(1 \cdot 12 - 1 \cdot 9.76) \cdot (1 - 1)$
 $= 11.428 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 30) (49.4 - 4.68) 1$
 $= 26.818 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot \pi r^2$$

$$= (2 \cdot \min(30, 300, 101)) \cdot (100 - 52.4) \cdot \pi$$

$$= 28.575 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.6786 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.7629 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.7629 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(12.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.68, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 52.375 = 45.828 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 56.0,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3461.9 kgf/cm ²	Passed
Occasional	: 60.1,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 11.6,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 45.8, tr = 4.68, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	34 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $25.49/52.01 = 0.490$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 47.6/0.5773$$

= 82.4893 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 766.2756 mm
 Parallel to Vessel Wall, opening length d 383.1378 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 30.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	33.251	33.251
Area Available in Shell	A1	5.714	5.714
Area Available in Nozzle Wall	A2	26.818	26.818
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	28.575	28.575
TOTAL AREA AVAILABLE	Atot	62.107	62.107

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: S2-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (49.9 - 11.4 + 2 \cdot 97 \cdot 1 \cdot (1 \cdot 12 - 9.76)) \cdot 1407)$
 $= 60204.70 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f r_2) \cdot S_v$
 $= (55.4 + 1 - 0 \cdot 1) \cdot 1407$
 $= 79351.90 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$
 $= (55.4 + 0 + 1 + (23.3)) \cdot 1407$
 $= 112109.83 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (55.4 + 0 + 1 + 0 + (23.3)) \cdot 1407$
 $= 112109.83 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 705 \cdot 10 \cdot 0.49 \cdot 1407$
 $= 76339. \text{ kgf}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 304) \cdot (100 - 3) \cdot 0.7 \cdot 1407$$

$$= 912266. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 705 \cdot (15 - 3) \cdot 0.74 \cdot 1407$$

$$= 138345. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (76339 + 912266) = 988604 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (76339 + 0 + 138345 + 0) = 214684 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (76339 + 138345 + 0) = 214684 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 988604 kgf, must exceed W = 60204 kgf	or W1 = 79351 kgf
Path 2-2 = 214684 kgf, must exceed W = 60204 kgf	or W2 = 112109 kgf
Path 3-3 = 214684 kgf, must exceed W = 60204 kgf	or W3 = 112109 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 29.9 kgf/cm²

The Drop for this Nozzle is : 134.1509 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 449.1509 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: S2-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1060.000	mm
Shell Thickness	15.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	504.850	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	25.49	kgf/cm ²
Radial Load	P	-3671.16	kgf
Circumferential Shear	Vc	3671.16	kgf
Longitudinal Shear	Vl	2753.37	kgf
Circumferential Moment	Mc	5.72	kgf-m.
Longitudinal Moment	Ml	5.72	kgf-m.
Torsional Moment	Mt	6.61	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.647) is > 0.5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (7.049) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 704.85 + 2 \cdot 1.65 \cdot \sqrt{(539.0(15.0 - 3.0))}$
 $= 970.249 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1078.000 mm
Nozzle Outside Diameter	(d) =	704.850 mm
Vessel Thickness used	(T) =	12.000 mm
Nozzle Thickness used	(t) =	97.000 mm
T / t	=	0.124
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.197$$

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.034
MrD/Mc	=	0.241
NrDL/Mc	=	0.090
M0d/Mc	=	0.069
N0DL/Mc	=	0.061
MrD/Ml	=	0.045
NrDL/Ml	=	0.038
M0D/Ml	=	0.014
N0DL/Ml	=	0.030

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	87	87	87	87
Outplane Bending (P)	309	-309	309	-309
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	1119	1145	1119	1145
Outplane Stress Summary	1516	923	1517	922

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	111	111	111	111



Inplane Bending (P)	1065	-1065	1065	-1065
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	559	559	559	559

Inplane Stress Summary	1734	-393	1738	-395
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	27	27	-27	-27
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	27	27	-27	-27
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1738	1317	1741	1319
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	87	87	87	87
Outplane Bending (P)	309	-309	309	-309
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-2	2	2	-2
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	559	559	559	559

Outplane Stress Summary	954	339	959	335
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	111	111	111	111
Inplane Bending (P)	1065	-1065	1065	-1065
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-8	8	8	-8
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1119	1145	1119	1145

Inplane Stress Summary	2287	198	2304	183
------------------------	------	-----	------	-----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-20	-20	20	20
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-20	-20	20	20
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
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Two * Max Shear Stress	2287	342	2305	338
------------------------	------	-----	------	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	87	87	87	87
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	67	98	67	98

Hoop Stress Summary	154	185	155	185
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	33	33	33	33

Axial Stress Summary	46	24	46	24
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	3	3	-3	-3
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	154	185	155	185
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	87	87	87	87
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	67	98	67	98

Hoop Stress Summary	154	185	155	185
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------



Axial Membrane (P)	1	1	1	1
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	33	33	33	33

Axial Stress Summary	46	24	47	24
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-2	-2	2	2
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	154	185	155	185

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1119	1145	1119	1145	1119	1145	1119	1145
Circ. Pl (SUS)	87	87	87	87	111	111	111	111
Circ. Q (SUS)	309	-309	309	-309	1057	-1057	1073	-1073
Long. Pm (SUS)	559	559	559	559	559	559	559	559
Long. Pl (SUS)	111	111	111	111	87	87	87	87
Long. Q (SUS)	1064	-1064	1066	-1066	307	-307	311	-311
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	27	27	-27	-27	-20	-20	20	20
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1119	1145	1119	1145	1119	1145	1119	1145
Pm+Pl (SUS)	1207	1233	1207	1233	1230	1256	1230	1256
Pm+Pl+Q (Total)	1737	1318	1739	1320	2287	341	2303	337

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1145.00	1407.13	Passed
Pm+Pl (SUS)	1256.66	2110.69	Passed
Pm+Pl+Q (TOTAL)	2303.30	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	67	98	67	98	67	98	67	98
Circ. Pl (SUS)	87	87	87	87	87	87	87	87
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	33	33	33	33	33	33	33	33
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	11	-11	11	-11	11	-11	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	67.0	98.0	67.0	98.0	67.0	98.0	67.0	98.0
Pm+Pl (SUS)	154.1	185.1	154.1	185.1	154.0	185.0	154.0	185.0
Pm+Pl+Q (Total)	154.1	185.1	154.1	185.1	154.0	185.0	154.0	185.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	98.00	1407.13	Passed
Pm+Pl (SUS)	185.06	2110.70	Passed
Pm+Pl+Q (TOTAL)	185.06	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	25.49	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	25.49	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	15.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1060.000	mm
Shell Temperature for Internal Pressure	Ts	180.00	°C
Shell Material		SA-516 70	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1407.13	kgf/cm ²
Shell Allowable Stress at Ambient		1407.13	kgf/cm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	20.39	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	20.39	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	14.0000	mm
Channel Internal Corrosion Allowance	cac	3.0000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1060.000	mm
Channel Design Temperature	TEMPC	85.00	°C
Channel Flange Matl		SA-516 70	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	1407.13	kgf/cm ²
Channel Allowable Stress at Ambient		1407.13	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	396	
Tube Wall Thickness	et	2.7700	mm
Tube Outside Diameter	D	25.4000	mm
Total Straight Tube Length	Lt	5250.00	mm
Straight Tube Length (bet. inner tubsht faces) L		5130.00	mm
Design Temperature of the Tubes		85.00	°C
Tube Material		SA-334 6	
Tube Material UNS Number		K03006	
Is this a Welded Tube		No	
Tube Material Specification used	Smls. & wld. tube		
Tube Allowable Stress at Temperature		1203.20	kgf/cm ²
Tube Allowable Stress At Ambient		1203.20	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2273.86	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	40.5000	mm
Tube Layout Pattern		Square	
Fillet Weld Leg	af	2.7700	mm
Groove Weld Leg	ag	2.7700	mm
Tube-Tubesheet Joint Weld Type	Full	Strength	
Method for Tube-Tubesheet Jt. Allowable		UW-20.3	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	517.036	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	6.0000	mm
Area of Tube Layout	Ap	6133.6	cm ²
Perimeter of Tube Layout	Cp	3085.1	mm



Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 85.00 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

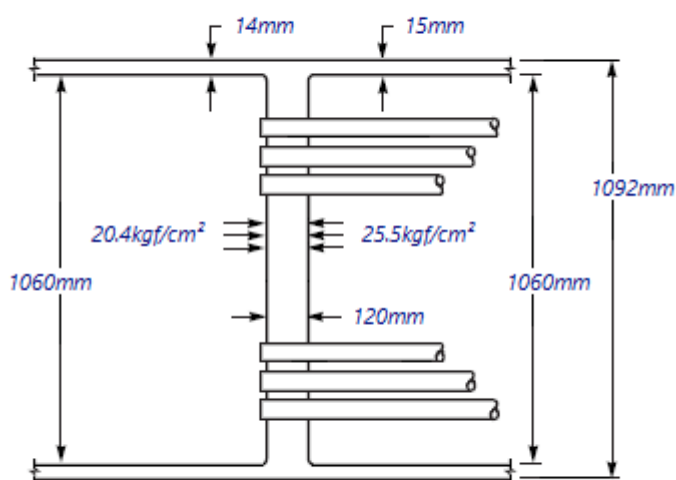
Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 120.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 1092.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	85.0 °C	0.20297E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%



Elastic Mod. at Tubsht. Design Temp. 85.0 °C 0.20297E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 85.0 °C 0.20297E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (20.4 \cdot (25.4/2 - 0)) / (1203 \cdot 1 + 0.4 \cdot 20.4)$
 $= 0.2138 + 0.0000 = 0.2138 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 85.00 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	25.40	5647.04	9.17	50.0000	0.0130823	1251.46

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 192 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6538	25.40	5647.04	38.85	50.0000	0.0007287	742.89

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 743) / (3 \cdot 38.9) = 25.5 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6538 mm
Allowable Internal Pressure at Corroded thickness	287.51 kgf/cm²
Required Internal Design Pressure	20.39 kgf/cm²
Allowable External Pressure at Corroded thickness	191.50 kgf/cm²
Required External Design Pressure	25.49 kgf/cm²
Required Thickness due to Shell Side pressure	0.6538 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 517 + 25.4$
 $= 1059.472 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (40.5 - 25.4) / 40.5$
 $= 0.373$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Ds / Do$ (Configurations a, b, c)
 $= 1060 / 1059$
 $= 1.0005$

Ratio [pc]:

$= Dc / Do$ (Configurations a, e, f)
 $= 1060 / 1059$
 $= 1.0005$



ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0/120$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 2.77 (2029713/2029713) \cdot$$

$$(1203/1407) \cdot (0); 25.4 - 2 \cdot 2.77)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 40.5 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1059 \cdot 40.5) / (\pi \cdot 1059^2)) }$$

$$= 40.5000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (40.5 - 25.4) / 40.5$$

$$= 0.3728$$

$$h/p = 2.962963 ; \mu^* = 0.372840 ; E^* = 865009 \text{ kgf/cm}^2$$

$$E^*/E = 0.426173 ; \nu^* = 0.319366$$

Note: As h/p (2.963) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ ((1060 + 15) 15) }$$

$$= 0.0143 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.014 \cdot 1973947 \cdot 15^3 / (6 (1 - 0.3^2))$$

$$= 17467166.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (1060 + 15) \cdot 17467166 / 120^3 \cdot (1 + 120 \cdot 0.014 + (120 \cdot 0.014)^2 / 2)$$

$$= 273402.2 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 1060^2 / (4 \cdot 1973947 \cdot 15) \cdot (1 - 1060 \cdot 0.3 / (2 (1060 + 15)))$$

$$= 0.80837\text{E-}02 \text{ mm/kgf/cm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1 \cdot 17467166 \cdot 0.014 \cdot 0.008083732 (1 + 120 \cdot 0.014)$$

$$= 5496.4604 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12 (1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ ((1060 + 14) 14) }$$

$$= 0.0148 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6 (1 - \nu_s^2))$$

$$= 0.015 \cdot 2029713 \cdot 14^3 / (6 (1 - 0.3^2))$$

$$= 15122227.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:



$$\begin{aligned}
 &= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (Cm\theta + h \cdot \beta c \cdot (Cm\Delta + Cq\theta)/2 + (h \cdot \beta C)^2/2 \cdot Cq\Delta) \\
 &= 6(1060 + 14/\cos(0)) \cdot 15122227/120^3 \cdot (1 + 120 \cdot 0.015 \cdot (1 + 1)/2 + (120 \cdot 0.015)^2/2 \cdot 1) \\
 &= 245951.6 \text{ kgf/cm}^2
 \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned}
 &= Dc^2/(4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc))) \\
 &= 1060^2/(4 \cdot 2029713 \cdot 14) \cdot (1 - 1060 \cdot 0.3/(2(1060 + 14))) \\
 &= 0.84218E-02 \text{ mm/kgf/cm}^2
 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$\begin{aligned}
 &= \rho c \cdot kc[\delta C(Cm\theta + \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \zeta c(Cm\theta + h \cdot Cq\theta \cdot h \cdot \beta C / 2)] \\
 &= 1 \cdot 15122227 [0.008421815 (1 + 0.015 \cdot 1 \cdot 120) - 0.000 (1/120 + 1 \cdot 0.015/2)] \\
 &= 5249.4331 \text{ mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / Do \\
 &= 1092/1059 \\
 &= 1.0307
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 &= (1 - v*)/E* \cdot (\lambda s + \lambda c + E \cdot \ln(K)) \\
 &= (1 - 0.32)/865010 \cdot (273402 + 245952 + 2029713 \cdot \ln(1.03)) \\
 &= 0.4570
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= Do^2/16 \cdot [(\rho s - 1) \cdot (\rho s^2 + 1) \cdot Ps - (\rho c - 1) \cdot (\rho c^2 + 1) \cdot Pt] \\
 &= 1059^2/16 \cdot [(1 - 1) \cdot (1^2 + 1) \cdot 25.5 - (1 - 1) \cdot (1^2 + 1) \cdot 20.4] \\
 &= 356.6575 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= Mts + \omega c \cdot Pt - \omega s \cdot Ps \\
 &= 357 + 5249 \cdot 20.4 - 5496 \cdot 25.5 \\
 &= -32706.2109 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\
 &= ((-32706) - 1059^2/32 \cdot 0.46 (25.5 - 20.4)) / (1 + 0.46) \\
 &= -78541.9062 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt) \\
 &= -78542 + 1059^2/64 \cdot (3 + 0.32) \cdot (25.5 - 20.4) \\
 &= 218292.0781 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| Mp |; | Mo |) \\
 &= \max(| -78542 |; | 218292 |) \\
 &= 218292.0781 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu* (h - hg')^2) \\
 &= 6 \cdot 218292 / (0.37 (120 - 6)^2) \\
 &= 270.3071 \text{ kgf/cm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 1407$$

$$= 2814.27 \text{ kgf/cm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 155278 / (0.37 \cdot (35.8 - 6)^2)$$

$$= 2809.1313 \text{ kgf/cm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc}$$

$$= 35.8 + 0 + 0$$

$$= 35.8252 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(35.8 ; 2.42)$$

$$= 35.8252 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |25.5 - 20.4|$$

$$= 5.099 \text{ kgf/cm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.37 \cdot 120 / 1059 \cdot \min(0.8 \cdot 1407 ; 0.533 \cdot 2341)$$

$$= 190.151 \text{ kgf/cm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.37 \cdot 1126) \cdot (4 \cdot 6134 / 3085) \cdot |25.5 - 20.4| + 0$$

$$= 2.4152 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 15.000mm

Min. Shell len. adjacent to the tubesheet is: 226.971 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1060^2 / (4 \cdot 15 (1060 + 15)) \cdot 25.5$$

$$= 444.1008 \text{ kgf/cm}^2$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu) / E \cdot$$

$$D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 385085 / 15^2 \cdot [0.014 \cdot 0.21 + 6(1 - 0.32) / 865010 \cdot$$

$$1059 / 120^3 \cdot (1 + 120 \cdot 0.014 / 2) \cdot (-78542 + 1059^2 / 32 \cdot$$

$$(25.5 - 20.4))]$$

$$= 1625.571 \text{ kgf/cm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 2069.672 \text{ kgf/cm}^2 \text{ (Should be } \leq 2110.70)$$

Note: For a given Channel thickness of: 14.000 mm

Min. Channel len. adjacent to the tubesheet is: 219.275mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1060^2 / (4 \cdot 14 (1060 + 14) \cdot \cos(0)) \cdot 20.4$$

$$= 381.012 \text{ kgf/cm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot [(\delta c \cdot Cm \Delta \cdot \beta c - \zeta c \cdot Cm \Theta) \cdot Pt - 6((1 - \nu) / E) \cdot Do / h^3 \cdot (Cm \Theta + h \cdot \beta c / 2 \cdot Cm \Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 333388 / 14^2 \cdot [(0.008421815 \cdot 1 \cdot 0.015 - 0.000 \cdot 1) \cdot 20.4 - 6((1 - 0.32) / 865010) \cdot 1059 / 120^3 \cdot (1 + 0.015 \cdot 120 / 2 \cdot 1) \cdot (-78542 + 1059^2 / 32 \cdot (25.5 - 20.4))]$$

$$= 924.792 \text{ kgf/cm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 1305.804 \text{ kgf/cm}^2 \text{ (Should be } \leq 2110.70)$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 1203$$

$$= 2369.474 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 1203$$

$$= 2369.4744 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min (S \text{ and } St)$$

$$= 1203 / 1203$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 1203$$

$$= 1569.6062 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 (\sqrt{ ((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd) - 0.75 \cdot do })$$

$$= 2 (\sqrt{ ((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1) - 0.75 \cdot 25.4 })$$

$$= 3.372 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm

Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 2369 is $\geq$ tube strength 2369 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	270.3 $\leq$	2814.3 kgf/cm ²	Ok
Tubesheet shear stress	22.7 $\leq$	1125.7 kgf/cm ²	Ok
Stress in Shell at Tubesheet	2069.7 $\leq$	2110.7 kgf/cm ²	Ok
Stress in Channel at Tubesheet	1305.8 $\leq$	2110.7 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
----------------	----------	--------	-----

Tubesheet Thickness :	35.825	120.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear Allwd	Case Type	Pass/ Fail
D1uc	72.433	...	1110	2814	91	1126 Ps+Pt D1	Ok
D2uc	82.472	...	1390	2814	113	1126 Ps+Pt D2	Ok
D3uc	35.825	...	270	2814	23	1126 Ps+Pt D3	Ok
D1c	78.097	...	1228	2814	94	1126 Ps+Pt-c D1	Ok
D2c	88.261	...	1536	2814	118	1126 Ps+Pt-c D2	Ok
D3c	36.694	...	299	2814	24	1126 Ps+Pt-c D3	Ok
Max:	88.2611	... mm	0.546	0.105	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	1049	2111	2619	4935 Ok
D2uc	...	...	3119	4638	1314	2111 Ok
D3uc	...	...	2070	2111	1306	2111 Ok
D1c	...	...	1257	2111	3266	4935 Ok
D2c	...	...	3850	4638	1568	2111 Ok
D3c	...	...	2593	4638	1698	2111 Ok
Max	...	...	0.981	0.805		

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	88.261	120.000	Ok
Tube thickness :	0.654	2.770	Ok
Tube-tubesheet fillet weld leg :	1.686	2.770	Ok
Tube-tubesheet groove weld leg :	1.686	2.770	Ok

Given Shell Thickness: 15.0000 mm
 Given Channel Thickness: 14.0000 mm

Min Shell length of thk, (15.000) adj. to tubesheet: 226.971 mm

Min Channel length of thk, (14.000) adj. to tubesheet: 219.275 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	47.218	1.000	47.417	1.000
Tubesheet Shear Stress	243.830	1.000	243.830	1.000
Shell Stress (Axial, Junction)	38.426	0.511	30.710	1.000
Tubesheet-Channel Junction	30.816	1.000	30.710	0.660
Tube Pressure Stress	287.514	1.000	191.501	1.000
Minimum MAWP	30.816		30.710	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	52.155	1.000	52.155	0.996
Tubesheet Shear Stress	253.330	1.000	253.330	1.000
Shell Stress (Axial, Junction)	52.155	0.518	42.913	1.000
Tubesheet-Channel Junction	41.351	1.000	42.913	0.843
Tube Pressure Stress	287.514	1.000	191.501	1.000
Minimum MAPnc	41.351		42.913	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(120/4, \min(120, 15)) \\
 &= 30.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(25.5/42.9, 1) \\
 &= 0.594
 \end{aligned}$$

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(120/4, \min(120, 14)) \\
 &= 30.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\
 &= \min(20.4/41.4, 1) \\
 &= 0.493
 \end{aligned}$$

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C





PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: